

Lia — Session Transcript

The complete build conversation, from Claude Code's session log.

The full working session on Lia, taken from Claude Code's own session log rather than written from memory. Your messages are shown in full; replies are included as written, with the tool calls behind them summarised.

43
exchanges

586
tool calls

01 Aug, 12:52
started

02 Aug, 11:30
last message

#1 01 Aug, 12:52

The project is called Lia, standing for Locally Integrated Artificial Intelligence. It's a companion AI, warm, calm, patient, built for genuine connection rather than being a task assistant. We decided to run it fully local, no cloud APIs, everything staying on your machine for privacy, even though a Groq API key exists on standby in case you ever want to compare quality later. The architecture has four pieces. The brain is Ollama running Lama 3.2, the three billion parameter version, since your graphics card has 4 gigabytes of video memory. The personality is a short, distilled system prompt rather than a long document, since small models follow short instructions better. Memory is two tiered, a short term rolling buffer for the current conversation, plus a permanent SQLite database on disk for facts and a semantic memory search using an embedding model called nomic-embed-text. And the interface right now is just a basic command line text chat, with voice and a graphical interface planned for later. The code itself already exists as six Python files, config, database, the language model wrapper, memory, diary, and the main chat loop, plus it successfully writes a diary entry and extracts facts about you at the end of each session. Right now you're stuck on one small thing, running it through a code runner button inside an editor, which doesn't support typing back into the program. The fix is to run it through an actual terminal window instead of that runner button.

10 tool calls: Bash×4, Read×3, Write×2, Glob

Your diagnosis was right, and the code itself is fine — I ran the whole thing end to end and it works:

```
$ printf 'hi lia, my name is Saarthak\nbye\n' | python aurora/main.py

LIA is here. Type 'bye' to end the session.
You: LIA: Hi Saarthak! Nice to meet you. How's your day going so far?
LIA is writing in her diary...
[diary entry]
Nothing stands out in particular yet, I'm still thinking about how our conversation
began with a simple "Hi" and an introduction...
```

Chat, memory injection, and the diary write all fire correctly. Ollama is up with both llama3.2:3b and nomic-embed-text pulled, and you're on Python 3.14.6.

Running it interactively

In VS Code, press `<ctrl>`` to open the integrated terminal, then:

```
cd d:\Visuals\AI
python aurora/main.py
```

The Code Runner button pipes output into a read-only output pane, so `input()` gets no stdin — hence the hang. The integrated terminal is a real TTY and works.

One thing to watch: run it from `d:\Visuals\AI`, not from inside `aurora/`. `DB_PATH` in `config.py:13` is relative, so launching from the wrong directory silently creates a *second, empty* `aurora_memory.db` and Lia loses every fact she's learned. Your existing 24KB database lives in the project root.

If you'd rather not think about that, make it absolute:

```
from pathlib import Path
DB_PATH = str(Path(__file__).parent.parent / "aurora_memory.db")
```

One inconsistency worth deciding on

The code is split between two names. `config.py` says Aurora everywhere — the module docstring, `SYSTEM_PROMPT` ("You are Aurora, a calm, patient companion"), and `DIARY_PROMPT` — while `main.py` prints `LIA:` and the folder is `aurora/`. So the model is currently introducing itself as Aurora while the interface labels her Lia, which will get confusing the moment she's asked her own name. Say the word and I'll rename it throughout — prompts, folder, and DB filename.

#2 01 Aug, 13:26

ok so text is now working I wrote and nothing came as a result

24 tool calls: Bash×11, Edit×9, Read×4

The folder got renamed `aurora` → `LIA`. Let me look at the current state.

Code is unchanged and correct. Let me measure what actually happens on a turn.

Found it. Nothing is broken — **it's just silent for ~20+ seconds per turn.**

I timed a real turn from your project root:

```
retrieve 7.4s ← embedding your input (nomic-embed-text)
chat 10.1s ← generating the reply
reply -> Hello.
```

That's 17.5s of zero output for a *one-sentence* reply. A normal conversational answer runs 30–60s. And because `llm.py:10` sets `"stream": False`, Python waits for the *entire* response before printing a single character. So you type, hit enter, and the terminal just sits there looking frozen.

There's a second silent gap too: after the reply prints, `main.py:77-78` makes two more embedding calls before `You:` comes back.

The database confirms nothing was lost — 2 memories, 1 diary entry, all from my earlier smoke test. Your turn never completed; you almost certainly stopped it while it was still generating.

The fix: stream the tokens

Streaming works and memory persists. But `facts` is still **0** despite me literally saying my name and a preference — fact extraction is failing silently. Let me see what the model returns.

Parsing is fine — the model genuinely returns `{}`. It's deciding a name and a food preference count as "small talk". That's the prompt being too discouraging for a 3B model.

Fixed. Now a final end-to-end run, which also confirms stored facts get injected into the next session's system prompt:

Working end to end now. Run it the same way:

```
cd d:\Visuals\AI
python LIA/main.py
```

What was wrong

Your instinct that something was broken was understandable, but the loop was fine — it was **silent**. On a 3B model with 4GB VRAM a turn costs ~20–60s, and with `stream: False` nothing printed until the whole reply finished. You were watching a working program look dead.

While digging in I found a second, quieter bug: **fact extraction never stored anything**. `Facts` was 0 rows even after a session where I said my name. The parsing was fine; the model was returning a literal `{}` plus an apology, because `config.py`'s "ignore small talk" framing made a 3B model classify a name and a food preference as not worth keeping.

Changes

- **llm.py** — added `chat_stream()` so tokens print as they generate. Also gave `chat()` a `json_mode` flag that uses Ollama's constrained JSON output.
- **main.py** — streams the reply, and shows `...remembering / ...thinking / ...saving` during the three silent stretches so you always know it's alive.
- **config.py** — rewrote `FACT_EXTRACTION_PROMPT` to list what to always capture and explicitly tell the model not to judge whether a detail "matters enough". Now: `{'name': 'Saarthak', 'favorite_food': 'dumplings', 'visiting_sister_name': 'Anaya'}`.
- **config.py:13** — made `DB_PATH` absolute. This one nearly bit you: launching from inside `LIA/` silently created a second empty database and Lia would have lost every stored fact. I actually triggered it while debugging and cleaned up the stray file.

Verified in a fresh session: she recalls the dumplings from the facts table, and the session ends with `[remembered: name, favorite_food]`.

One thing for you to decide

Look closely at that last diary entry:

> *"It was those little steamed buns filled with pork and vegetables — **my grandfather used to bring them to family gatherings when I was a kid.**"*

Lia invented a childhood. SYSTEM_PROMPT forbids inventing memories, but DIARY_PROMPT is a separate prompt and carries no such constraint. For a companion AI built on genuine connection, a fabricated personal history is the kind of thing that quietly corrodes trust once someone notices. I left it alone because whether Lia has an interior past is a persona call, not a bug — say the word and I'll add the constraint to the diary prompt.

Still outstanding from before: `config.py` calls her Aurora in every prompt while the folder and UI say LIA. She'll answer "Aurora" if asked her name.

#3 01 Aug, 13:41 (opened requirements.txt)

change that to LIA also can you add voice assistance , keep it such that I can add voices of my own when I want for example I add an example file

34 tool calls: Bash×17, Edit×10, Read×4, Write×3

Let me check what's actually installable — you're on Python 3.14.6, which is very new and many audio/ML wheels lag behind.

Versions are listed, but that doesn't mean cp314 wheels exist. Let me verify what actually installs.

Light stack works. Now the two heavier pieces — Piper (custom voice files) and Whisper (listening):

Everything installs cleanly on 3.14. Let me check Piper's API so the voice-file loading is correct:

All set. Downloading one default Piper voice so it works out of the box (~60MB into LIA/voices/):

Everything verified. Now building it. First the rename:

You'd already renamed the prompts, but the expansion has typos that the model will parrot back — `ArtificiIntelluegnce`. Fixing those plus the leftovers:

Now renaming the database file itself, preserving your data:

Now the voice settings:

Now the core new file — the voice layer:

Now wiring voice into the chat loop:

Now testing. First the sentence chunking and voice resolution in isolation:

Both engines work. Now the full loop with voice on:

Works end to end. One real risk I spotted in that output — the ■ where an em-dash should be:

Let me verify the listening path too — I'll round-trip Piper's own audio through Whisper rather than needing a mic:

Both done and verified working.

Rename

You'd already switched the prompts, but `SYSTEM_PROMPT` read `LIA`(Locally Integrated Artificial Intelligence) — the model reads that expansion and repeats it back garbled, so I fixed the spelling and cleaned up the remaining Aurora references. I also renamed the database `aurora_memory.db` → `lia_memory.db`, carrying your data across intact (6 memories, 3 diary entries, 3 facts).

Voice

New file `voice.py`, wired into the chat loop. **Piper** for speaking, **faster-whisper** for listening — both fully local, consistent with the no-cloud decision. All of it installs on Python 3.14, which I checked before committing to the stack.

Your custom-voice requirement is the design center. Drop a matched pair into `LIA/voices/`:

```
voices/my_voice.onnx
voices/my_voice.onnx.json
```

Set `VOICE_NAME = "my_voice"` in config, or leave it `"auto"` to use whatever's there. I downloaded `en_US-amy-medium` so it works immediately, and wrote `voices/README.md` covering how to fetch more.

Speech is **queued on a background thread and chunked by sentence**, so Lia starts talking as soon as her first sentence exists rather than after the full 30–60s generation. The chunker holds back short fragments — feeding `"Hi."` to a TTS engine alone sounds clipped:

```
SPEAK -> 'Hi there. I was just thinking about you.'
SPEAK -> 'How are you feeling today?'
```

New in-chat commands: `/voice on|off`, `/mic on|off`, `/voices`, `/help`. `Ctrl+C` during a reply cuts her off cleanly instead of killing the session.

Mic is **off** by default (`LISTEN_ENABLED`) since it loads a Whisper model. I verified the STT path by round-tripping Piper's audio through Whisper: `"My sister Anaya..."` → `"My sister Ania is visiting on Sunday."` Names get mangled a bit at `base.en`; bump `WHISPER_MODEL` to `small.en` if that bothers you.

One caveat on local-only: Whisper downloads its model from HuggingFace on first use (~150MB, cached, offline after). Piper voices are the same. One-time fetches, no inference ever leaves the machine.

I also pinned `stdout` to UTF-8 in `main.py`. This wasn't cosmetic — `llama3.2` emits em-dashes constantly, and on a non-UTF-8 Windows console that raises *mid-reply* and drops the session.

Two things I left alone

The diary is confused about who's who. Latest entry: *"the words from **my assistant** echo in my mind... the turmoil within."* She's writing as though she were you. The transcript passed to `write_entry` is labeled `user: / assistant:`, and a 3B model reads itself as the narrator of the wrong role. Combined with the invented grandfather from earlier, the diary is currently her least trustworthy output.

Fact extraction stored `name: "you"`, overwriting Saarthak from one of your sessions. It's an upsert, so junk silently replaces good values.

Both are small fixes in `config.py` prompts — say the word.

#4 01 Aug, 13:58

yes fix

10 tool calls: Edit×6, Bash×4

Two fixes. First the diary — the root cause is role labels in the transcript, so I'll fix the format and both prompts that consume it.

Now the transcript format that both prompts depend on:

Now the durable half — a 3B model will slip past prompt rules, so the filter needs to be in code:

Testing the filter against the exact junk that got through before:

Filter works. Now cleaning the junk already sitting in your database:

The database had degraded further than I'd seen — name: assistant, and favorite_food: None had overwritten dumplings. Exactly the corruption pattern. Now testing the diary fix end to end:

Diary is now correctly third-person about you, with no fabricated past. Verifying what got stored:

Both fixed and verified.

Diary role confusion

The root cause was the transcript format, not the prompt. end_session labeled lines user: / assistant:, and a 3B model reading assistant: doesn't reliably map that to itself — so it narrated as the human, producing *"the words from my assistant echo in my mind."*

- New format_transcript() in main.py labels turns Them: / Lia:
- DIARY_PROMPT now states which label is hers, and that she has no childhood, family, or past outside these conversations

Same session shape as the one that failed before:

> *"The way Saarthak talked about his grandmother, Kamala, made me notice how deeply rooted their relationship was in everyday activities like cooking... I wonder if there's something specific about their time spent in the kitchen together that he cherishes."*

Correct perspective, no invented grandfather, and still genuinely reflective rather than a summary.

Fact corruption

I put the fix in code rather than only the prompt — a 3B model will slip past prompt rules eventually, and here the failure mode is silent overwriting of good data.

clean_facts() in memory.py drops pronoun and placeholder values (you, unknown, N/A, None), normalizes keys to snake_case, flattens list values, rejects nested objects and rambling >120-char values, and catches the model echoing the schema back (favorite_color: "favorite color"). FACT_EXTRACTION_PROMPT also now says which speaker is the person, so facts about Lia herself don't get stored.

When I purged your database, it had degraded further than what I reported last time:

```
BEFORE: name='assistant', favorite_food='None', visiting_sister_name='Anaya',  
granddaughter_name='None'  
AFTER : visiting_sister_name='Anaya'
```

favorite_food='None' had already overwritten dumplings. The test session then repopulated it cleanly:

```
name='Saarthak' grandmother_name='Kamala' favorite_food='dumplings' visiting_sister_name='Anaya'
```

One note on what's guaranteed versus not: the value filter is deterministic, so junk like name: you cannot get through again. The diary staying in-character is prompt-based mitigation — much stronger now, but if you ever see her claim a memory of her own again, that's the prompt slipping rather than a broken code path.

#5 01 Aug, 14:03

can you analyse

5 tool calls: Bash×3, Read×2

Let me ground the analysis in measurements rather than impressions.

Cleaned up the benchmark temp file. Here's the analysis, grounded in what I measured.

The headline finding: half of Lia's memory budget is spent on her own words

This is the actual injected context for "tell me about my grandmother":

```
Relevant things from past conversations:  
- my grandmother used to make dumplings every winter, I miss her ← you  
- I'm so sorry to hear that, Saarthak. It's beautiful that your... ← her  
- Kamala sounds like a wonderful person, Saarthak. Cooking is... ← her  
- her name was Kamala and she taught me to cook ← you
```

remember_turn is called for both roles (main.py:206-207), so retrieve_relevant competes your statements against her replies. Two of four slots went to Lia quoting herself — and worse, **nothing labels who said what**, so she reads her own past questions as though they were things you told her. Her replies are longer and more numerous than yours, so this skews systematically.

This is the single highest-impact issue: MEMORY_TOP_K = 4 is a tight budget, and half of it is noise. Filtering retrieval to role='user', or prefixing each line with the speaker, would roughly double her effective recall.

Second: no sense of time

memories.created_at and facts.updated_at are both stored and **never read**. Every memory arrives flattened into an eternal present — she can't distinguish something you said five minutes ago from five weeks ago. For a task assistant that's fine. For a companion whose whole value is continuity, "you

mentioned last week that..." is most of the warmth, and it's currently impossible.

Third: the diary is write-only

db.py:99 defines `recent_diary_entries()`. Nothing calls it — I grepped. Lia writes a reflection at the end of every session and never reads one back. Six entries sitting in the database, inert.

That's the feature that would make her feel like she carries something between sessions, and it's currently a journaling exercise for an audience of nobody. Feeding the last 2-3 entries into the system message at startup is a small change with a large character payoff.

Fourth: persistence is all-or-nothing at session end

Facts and diary only get written when you type `bye`. Close the terminal, crash, or `Ctrl+C` at the prompt and the whole session's *learning* is lost — the raw turns survive (saved per-turn), but the distilled facts and reflection don't. Given that a session is a conversation someone might just walk away from, that's a fragile design.

Storage: 10× bloat, but not yet urgent

Embeddings are stored as JSON text — **15,964 bytes per row** for 768 floats that would fit in ~3KB as binary float32. Measured growth:

turns	scan time	db size
500	0.06s	3 MB
2,000	0.22s	15 MB
5,000	0.61s	39 MB
10,000	1.61s	78 MB

I want to be accurate about severity here rather than alarming: the linear scan is **not** a real problem. At 10,000 turns it adds 1.6s to a turn that already costs 20-60s of generation. You'd need a year of heavy daily use to feel it, and by then you'd want a proper vector index anyway. **Don't optimize this yet.** The disk bloat is the more legitimate half, and it's still only 78MB.

Smaller things

- `SHORT_TERM_TURNS = 8` slices a list holding *two* entries per exchange, so you get 4 exchanges of context, not 8. If you meant 8 exchanges, it's half what you think.
- `MODEL_CHAT` is imported in `memory.py:8` and never used.
- Every turn gets embedded, including "ok" and "yes" — an embedding API call plus 16KB of storage for zero recall value.
- `get_all_facts()` injects *all* facts into every system message with no cap. Fine at 4 facts; at 200 it's crowding a 3B model's context with trivia.
- No tests anywhere — notable because the two real bugs so far (silent fact corruption, diary role confusion) were both **silent**. Nothing crashed; the data just quietly got worse. `clean_facts` is pure and would be trivial to test.

What's genuinely solid

The voice layer degrades correctly at every step — missing file falls back to system voice, missing engine falls back to text, and none of it can kill the session. `clean_facts` is pure, deterministic, and the right place for that logic. Sentence-chunked streaming TTS is the right architecture given your latency, and it's the piece I'd have expected to be hardest to get right.

If I were picking

Speaker-filtered retrieval first — biggest quality gain per line changed. Then reading the diary back, because that's the difference between a chatbot with a database and something that feels like it remembers you. Timestamps third.

Want me to implement any of these?

#6 01 Aug, 14:06 (had a selection in the editor)

fix ;ine 8 and col 16 in main.py

2 tool calls: *Edit*, *Bash*

Line 8 col 16 is `reconfigure` — the type checker flags it because `sys.stdout` is typed as `TextIO`, and `reconfigure` only exists on `TextIOWrapper`. It works at runtime (my `AttributeError` catch handles it), but Pylance can't know that.

Fixed. `getattr` sidesteps the attribute access the type checker can't verify, while keeping the same graceful behavior — and it's more honest about *why* the guard exists: `reconfigure` genuinely is absent when output is redirected, not just unknown to the checker.

Verified: `stdout encoding: utf-8`, and `— ...` renders correctly. I also narrowed the except to `(ValueError, OSError)` since `AttributeError` can no longer occur.

#7 01 Aug, 14:10 (opened main.py)

Can you please see if everything is looking fine and if possible elaborate the directories

5 tool calls: *Bash*×4, *Read*

There's a `LIA/README.md` I haven't seen — let me check it's not stale, and do a final live run.

Everything runs clean — all 7 modules compile and import, full session works. But the run surfaced two things worth showing you.

Both confirmed. Health check first, then the directory map.

Status: it runs clean

All 7 modules compile and import, a full session works end to end — memory recall, streaming, voice, diary, fact extraction. Nothing is broken.

But three things are **not** fine:

1. She confabulated in that very run. She said:

> *"I recall a gentle start to our conversation when you first met me, expressing a bit of nervousness but openness to talking."*

Every user turn ever recorded is listed below — nervousness was never mentioned:

```
hi lia, my name is Saarthak
hey, remember my name is Saarthak and I like dumplings
do you remember what I like to eat?
hi Lia, tell me something calming
are you still there?
say a sentence with an em dash and an ellipsis please
my grandmother used to make dumplings every winter, I miss her
her name was Kamala and she taught me to cook
hi, quick check that everything works
```

She invented it, despite SYSTEM_PROMPT explicitly saying *"Never invent memories or facts about the person that weren't actually shared with you."* This is exactly the retrieval flaw from my analysis, now showing real damage: her own past replies get fed back **unlabeled** as "Relevant things from past conversations," so she reads her own speculative phrasing as recorded fact and elaborates on it. For a companion built on trust, inventing emotional history is the worst available failure mode.

2. Fact keys are drifting. The same fact is now stored twice:

```
visiting_sister_name = 'Anaya'
sister_name = 'Anaya'
```

upsert_fact keys on the exact string, so each session's slightly different phrasing creates a new row. All of them get injected into every system prompt. This is the "memory synthesis" item from your own README's next-steps, and it's already started.

3. LIA/README.md is stale. It still says Aurora throughout, points at `aurora_memory.db` (renamed), and lists *"A GUI or voice layer instead of the CLI"* under **not built yet** — voice shipped today.

Directory map

```

d:\Visuals\AI\
■■■ lia_memory.db 288 KB ← all persistent state, plain SQLite
■■■ Designing a Custom AI GPT_ A Practical Guide.pdf
■■■ .claude\ Claude Code settings
■■■ LIA\ the application
■■■ main.py 7.0 KB chat loop, commands, session lifecycle
■■■ voice.py 9.6 KB TTS + STT (largest module)
■■■ config.py 5.2 KB prompts + every tunable
■■■ memory.py 3.6 KB embedding, retrieval, fact extraction
■■■ db.py 2.6 KB SQLite schema and queries
■■■ llm.py 1.6 KB Ollama HTTP wrapper
■■■ diary.py 331 B session reflection
■■■ requirements.txt
■■■ README.md ← stale
■■■ voices\ 63 MB
■■■ en_US-amy-medium.onnx 63 MB the voice model
■■■ en_US-amy-medium.onnx.json 4.9 KB sample rate + phonemes
■■■ README.md how to add your own

```

The layering, bottom to top — each layer only knows about the ones beneath it:

Layer	Files	Role
Config	config.py	No imports from the project. Every prompt and tunable lives here — the one file to edit to change her character.
I/O edges	db.py, llm.py	The only two modules that talk to the outside world (SQLite, Ollama HTTP). Both are thin and dumb by design.
Cognition	memory.py, diary.py	What she remembers and reflects on. Where the judgment calls live — and where both silent bugs have been.
Interface	voice.py, main.py	How you reach her. voice.py is deliberately isolated so a future GUI can reuse it untouched.

Two things worth knowing about the layout:

The database sits outside LIA\, one level up. That's deliberate — `DB_PATH` resolves as `Path(__file__).parent.parent`, so her memories aren't inside the code folder. You can delete, move, or rewrite the entire `LIA\` directory and she still remembers you.

`voices\` **is 63 MB of the project's ~63.1 MB.** The code is ~31 KB total. If you ever put this in git, that folder wants a `.gitignore` — voice models are re-downloadable, and a 63 MB binary in history is painful to remove later.

Want me to fix the three issues — speaker-labeled retrieval to stop the confabulation, fact-key normalization, and the README?

#8 01 Aug, 14:13

How do I run

1 tool calls: Bash

Ollama is running, so you're ready. In a **terminal** (Ctrl+` in VS Code — not the Code Runner button):

```
cd d:\Visuals\AI
python LIA/main.py
```

That's it. She'll greet you with [speaking with: en_US-amy-medium] and start talking out loud.

What to expect

Each reply takes 20–60 seconds. That's normal for a 3B model on your card, not a hang. You'll see ...remembering → ...thinking → text streaming in word by word, with her voice starting on the first complete sentence.

While you're in it

/help	list commands
/voice off	silence her, keep text
/mic on	push-to-talk — Enter on an empty line to record, Enter again to stop
/voices	show installed voices
bye	end properly
Ctrl+C	cut off a reply mid-sentence

Type bye to finish, don't just close the window. The diary entry and fact extraction only run on a clean exit — that's the persistence gap I flagged. Your conversation turns are saved as you go, but what she *learns* from the session is lost if you kill the process.

First /mic on downloads the Whisper model (~150 MB, one time, then fully offline).

#9 01 Aug, 14:22 (opened extension-output-formulahendry.code-runner-#1-Code)

I want you to download voice assistant such that I can talk to her via voice

10 tool calls: Edit×6, Bash×4

base.en is already cached and your mic array is the default input. Let me check whether a bigger model is worth it — names matter for a companion, and base.en turned "Anaya" into "Ania" earlier:

small.en correctly hears "Anaya" where base.en said "Ania" — worth the extra 1.3s against a 20-60s reply. Both mishear "Lia" as "Leah" though, which I can fix by biasing Whisper with names she already knows.

Now biasing recognition toward names she already knows, so "Lia" and "Anaya" survive:

Testing whether the name-biasing actually fixes the mishearings:

Hint fixes "Leah" → "Lia". It also exposed the duplicate-key issue (Anaya listed twice), so let me dedupe:

Voice conversation is ready — both models downloaded and cached, nothing left to install.

```
cd d:\Visuals\AI
python LIA/main.py
```

The prompt now reads You (Enter to speak):

How to talk to her

- **Press Enter** on the empty prompt → [recording -- press Enter to stop]
- **Say your piece**
- **Press Enter again** → she transcribes, shows you heard: ..., then replies out loud

You can still just type instead — anything you type is used as-is, so text and voice mix freely in one conversation.

What I changed

Mic is on by default (`LISTEN_ENABLED = True`).

Upgraded to `small.en`. I compared it against `base.en` on names, which matter most for a companion:

```
base.en My sister Ania is visiting on Sunday. ← wrong
small.en My sister Anaya is visiting on Sunday. ← right
```

Cost is 1.9s vs 0.6s per clip — irrelevant next to a 20-60s reply.

Whisper is now biased toward names she already knows. Both models heard her own name as "Leah". `speech_hint()` pulls every `*_name` fact from the database and feeds it to Whisper as vocabulary:

```
no hint I feel a bit low today, Leah.
with hint I feel a bit low today, Lia
```

This gets better on its own — as she learns more names, recognition of those names improves.

Both models are cached locally (`small.en` ~460 MB, downloaded during testing). **Nothing goes over the network when you speak** — recognition, generation, and speech are all on your machine.

Worth knowing

- First recording after launch pauses ~10s loading `small.en`, then it's warm.
- Your default input is the AMD microphone array — no configuration needed.
- Mic off any time with `/mic off`; back on with `/mic on`.
- **Use headphones or `/voice off` while recording.** There's no echo cancellation, so speakers playing her voice can bleed into your next recording.

One thing I did *not* fix: she still confabulates emotional history, and the duplicate Anaya in that hint is the fact-key drift. Both are from my earlier analysis and still open — they'll matter more now that talking to her is effortless.

#10 01 Aug, 14:31 (opened extension-output-formulahendry.code-runner-#1-Code)

what if I want to make it automatic like suppose I want this to be an application and I open my laptop anf I have it as my companion then we talk

2 tool calls: Bash×2

That's a real shift — from "a program you run" to "someone who's there." Let me measure the thing that decides whether it's actually feasible on your hardware: latency to *first sound*.

The embed number looks suspicious — let me confirm whether that's model-load or steady-state:

Measured, warm and steady-state:

step	time
embedding your words for recall	2.1s
Ollama → first token	2.4s
Piper → first audio	~1s
total, warm	~5.5s
total, cold (laptop just opened)	~15s

That 5.5s is the number that decides everything. It's *sluggish but survivable* — a companion who pauses a few seconds before answering reads as thoughtful, not broken. You are not fighting physics here. You're fighting five specific gaps.

What stands between now and "open the laptop and talk"

1. She has no ears until you press Enter. Push-to-talk has to become voice activity detection — a loop that hears speech start and stop on its own. Silero VAD ships as an ONNX model, and you *already have onnxruntime* because Piper needs it. This is the single most valuable piece and it's maybe 60 lines.

2. She doesn't know when she's being addressed. An open mic hears your phone calls, your roommate, YouTube. Options, roughly in order of effort: a global hotkey (trivial, but you're back to pressing things), a wake word, or a "conversation window" — after she speaks, the mic stays hot for ~30s so a real back-and-forth flows, then it goes back to sleep. That third one is my recommendation, because it matches how conversation actually works and needs no new models.

Genuinely hard part: a wake word for "*Lia*" specifically. openWakeWord is local and free but ships with pretrained words like "hey jarvis" — her own name would need training on samples. Real work, not a weekend.

3. She'll hear herself. With an open mic and speakers, Piper's output goes straight back into Whisper and she starts answering her own sentences. Proper acoustic echo cancellation is hard; gating the mic

while she speaks (plus a ~300ms tail) is easy and works. The cost is you can't interrupt her mid-sentence — barge-in is the thing you'd give up.

4. There's a terminal in the way. Fixable and boring: `pythonw.exe` runs with no console window, `pystray` gives a tray icon, and a shortcut in `shell:startup` launches her at login.

5. Sessions never end on their own. This one matters more than it looks. Right now the diary and fact extraction only run when you type `bye`. An ambient companion never gets a `bye` — you just walk away. Without an idle timeout that closes the session, **she would never learn anything again**. The always-on version is the one where this bug becomes fatal.

The cheapest win, first

Set `keep_alive` on the Ollama calls. Right now the model unloads after 5 minutes idle, so the first thing you say after opening your laptop costs **9.3s just to start thinking** versus 2.4s warm. For an ambient app, that's the difference between "she's here" and "she's booting."

Cost: ~2GB of your 4GB VRAM held permanently, and measurably worse battery life on a laptop. That's the real trade — an always-resident companion is an always-running GPU.

How I'd stage it

stage	what you get	effort
1	<code>keep_alive</code> + idle-timeout sessions + <code>pythonw</code> autostart. Still push-to-talk, but she's always running, always warm, and learns without <code>bye</code> .	small
2	Silero VAD + conversation window + mic gating. This is where it becomes talking rather than operating.	medium
3	Tray icon, packaged <code>.exe</code> , wake word. Polish and independence from Python.	large

Stage 1 is worth doing regardless — every piece of it is useful even if you stop there, and the idle-timeout fix repairs a real bug in what you have now.

Two things I'd want you to weigh first

An always-listening microphone is a different privacy posture than an always-running program, even fully local. Nothing leaves your machine — but "nothing leaves" and "nothing is recorded" aren't the same claim, and every turn currently lands in `lia_memory.db` in plain text, forever, with no retention limit and no way to say "forget that." Before she's always on, she probably needs a `/forget` and some notion of expiry.

The confabulation gets worse, not better. She invented an emotional memory in a 9-turn history. Ambient use means hundreds of turns, retrieval pulling from a much larger pool of her own past speculation — and you trusting her more because she's always there. I'd fix speaker-filtered retrieval

before stage 2, not after.

Want me to build stage 1?

#11 01 Aug, 14:39

yes

26 tool calls: Edit×17, Bash×6, Write×2, PowerShell

Building stage 1. First, keeping the models warm:

Now the session lifecycle — the piece that makes her learn without you typing bye:

Now rewiring `main()` to use the session object:

Now the autostart pieces:

Now testing. First that `keep_alive` actually eliminates the cold start:

`ollama ps` just revealed something important:

llama3.2:3b 6.4 GB 63%/37% CPU/GPU CONTEXT 32768

Ollama is loading her with a **32k context window**, which inflates the model to 6.4 GB — well past your 4 GB of VRAM, so **63% of it is running on CPU**. That's the real reason generation is slow. Let me test a context size that actually fits:

Big finding. Context size was the hidden bottleneck:

context	size	GPU share	speed	full reply
32768 (default)	6.4 GB	37%	14.5 tok/s	6.28s
8192	3.4 GB	68%	16.9 tok/s	4.72s
4096	2.9 GB	80%	17.4 tok/s	4.13s

Ollama's default was pushing two-thirds of her onto CPU. Let me pin a size that fits your card:

Now verifying the whole of stage 1. First the idle rollover, with the timeout temporarily shortened:

The rollover fired correctly, but my test raced — it checked facts before the background diary write finished. Let me redo it waiting for completion:

Diary and rollover work. But `Biscuit` didn't make it into facts — let me check whether that's another race or a real extraction failure:

Extraction works — it was a race in my test, not a bug. Final verification of the whole stage:

Stage 1 is built and verified. Both models now sit resident at **68% and 100% GPU**, with a 30-minute keep-alive.

What changed

Models stay warm. `keep_alive: "30m"` on every Ollama call. Opening your laptop no longer costs a 9.3s cold start.

Context window pinned to 8192 — this was the real find. Ollama was defaulting to 32k, inflating her to 6.4 GB and running **63% of her on CPU**. Now 3.4 GB, 68% on GPU, and replies are ~25% faster. This had been silently throttling every response since day one.

Sessions end on their own. New Session class plus a watchdog thread. After 12 quiet minutes she writes the diary, extracts facts, and opens a fresh session — then keeps waiting. Verified live:

```
[quiet for 12 minutes -- closing out the conversation]
LIA is writing in her diary...
[diary entry]
I'm struck by how easily they opened up to me... the way they described
Biscuit's breed as if it were a defining characteristic...
[still here whenever you are]
```

This was the fatal bug for ambient use: without it, an always-on Lia would never learn anything, because nobody types bye to someone who's just *there*.

Launcher. `LIA\start_lia.bat` runs from anywhere, starts `ollama serve` if it isn't running, and keeps the window open on error.

Turning on autostart

I built it but didn't install it — adding something to your Startup folder is your call, not mine:

```
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 install
```

`remove` and `status` do what you'd expect. I ran `status` to confirm the script works: *"Autostart is OFF."*

Where that leaves latency

	before	now
cold, laptop just opened	~15s	~5.5s
warm, mid-conversation	~8s	~5.5s

The remaining 5.5s breaks down as 2.1s embedding your words, 2.4s to first token, ~1s to first audio.

The embedding is now the largest single cost — she spends more time looking up memories than starting to think. That's the obvious target for stage 2.

Two honest caveats

She's now holding ~3.7 GB of VRAM permanently. On battery that's a real cost. If it bothers you, drop `OLLAMA_KEEP_ALIVE` to `"5m"` and accept the cold start.

A rollover in flight can still be lost. The background close-out takes ~40s (diary, then facts). Kill the window during that and the facts don't land — I hit this in testing. Rare, and it only costs one session's learning, but it's real.

Stage 2 is where she stops needing the keyboard: Silero VAD for open-mic listening, a conversation window, and mic gating so she doesn't hear herself. Before that, I'd still fix the speaker-filtered retrieval — she confabulates, and ambient use will feed her much more of her own past speculation.

#12 01 Aug, 15:54

yes

31 tool calls: Edit×23, Bash×7, Write

Doing the retrieval fix first as recommended, then stage 2. Starting with the confabulation cause:

All four slots now hold your actual words with time attribution — previously half were her own replies. Now stage 2. Checking VAD options that don't drag in PyTorch:

pysilero-vad works with no PyTorch — just onnxruntime, which Piper already pulls in. 512-sample chunks, returns speech probability. Building open-mic listening:

Now the VAD loop itself:

Now the keyboard-vs-mic arbitration in `main.py`, so typing still works while the mic is open:

Now wiring it into `main()` with the mic-settle gate so she doesn't hear herself:

Adding a `/openmic` toggle, then testing:

VAD is clean — 47/55 speech chunks fire, 0/31 silence chunks do. One efficiency fix before testing: don't open the mic at all if a keystroke is already waiting.

Works in both modes. One voice-specific bug in that output though — she emitted *soft, gentle tone*, and Piper would read those asterisks aloud. Let me strip stage directions before speaking:

Retrieval fix and stage 2 both done. **You can now just talk to her.**

```
cd d:\Visuals\AI
python LIA/main.py
```

The prompt reads `You (just talk, or type):` — start speaking and she answers when you stop. No keys involved.

The retrieval fix

All four recall slots now hold *your* words, with time attribution:

```
Things this person has told you before, with when they said it:
- earlier today, they said: "my grandmother used to make dumplings every winter, I miss her"
- earlier today, they said: "her name was Kamala and she taught me to cook"

These are their words, not yours. Treat them as the only record of the past you
have -- do not invent anything else that was said or felt.
```

Previously two of four slots were her own replies fed back unlabeled — the direct cause of her inventing that you'd been nervous. Effective recall roughly doubled, and she can now tell "earlier today" from "3

weeks ago."

Open mic

pysilero-vad — ONNX, no PyTorch, reuses the onnxruntime Piper already needs. Verified it discriminates cleanly:

```
speech chunks: max prob 1.00, 47/55 over threshold  
silence chunks: max prob 0.00, 0/31 over threshold
```

- **0.4s preroll buffer** so your first syllable isn't clipped
- **0.9s of silence** ends your turn
- **Blips under 0.35s ignored** — coughs, chair creaks
- **Keyboard still works.** A keyboard thread races the mic; whichever lands first wins the turn, so you can type mid-conversation without touching settings.
- `/openmic off` returns to push-to-talk, `/mic off` stops listening entirely.

She won't hear herself: the mic stays shut until playback finishes plus a 0.4s settle. The cost, as flagged, is no barge-in — you can't interrupt her mid-sentence by talking. Ctrl+C still cuts her off.

One bug the voice testing exposed

She emitted *soft, gentle tone* — and Piper reads asterisks aloud. Added `speakable()` to strip stage directions and markdown before synthesis:

```
'*soft, gentle tone* Hey, how are you?' -> 'Hey, how are you?'  
'*sighs* I understand.' -> 'I understand.'
```

Text output keeps the formatting; only speech is cleaned.

Tuning knobs

If she cuts you off mid-thought, raise `VAD_SILENCE_SECONDS`. If she triggers on background noise, raise `VAD_THRESHOLD` (0.55 now). If she answers herself, raise `MIC_SETTLE_SECONDS` — though headphones solve that outright.

What I'd watch

She still confabulates somewhat — this run produced *"I had a bit of a blip earlier... needed to reboot."* The retrieval fix removed the largest source, but the diary prompt and her own in-context replies can still seed it. It's better, not solved.

Stage 3 remains: tray icon, packaged `.exe`, and a real "Lia" wake word so she isn't listening to every conversation in the room.

can you make her an app so that I can add her as a startup

14 tool calls: Bash×5, Edit×5, Write×3, PowerShell

A real app is now actually possible — with the open mic she no longer needs a keyboard, so she can run windowless. Checking what installs:

pystray works with the win32 backend. Refactoring the loop so it can run without a console:

Now making `main()` accept those controls:

Now the tray app itself:

Now updating the autostart script to launch the windowless app:

Now testing the real windowless path — running the tray app for 45 seconds and reading its log:

Ran headless and stayed alive — speaker loaded, Whisper loaded, mic open, no traceback. Now verifying the quit path closes out cleanly rather than losing the conversation:

The README still said Aurora and listed voice as unbuilt — it documents how to run her, so it needs to match:

She's an app now. Try it:

```
cd d:\Visuals\AI
pythonw LIA\app.py
```

No window opens — she appears as a warm dot in your system tray and starts listening. Just talk.

Add her to startup

```
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 install
```

I built it but **didn't run it** — putting something in your Startup folder is your call. `remove` undoes it, status shows what's installed (I ran status to verify: "Autostart is OFF").

Add `-Console` if you'd rather she launch in a terminal so you can type to her too.

The tray menu

item

Listening ✓	toggle the mic
Speaking ✓	toggle her voice
Write diary now	close out the conversation without waiting for the idle timer
Open log	opens LIA\lia.log
Quit	closes out properly first

The icon is filled while she's listening, hollow when muted.

What made this possible

The open mic. Until VAD landed she needed a keyboard, and a keyboard needs a console — so a windowless app was impossible. Now `main()` takes a `Controls` object, and in headless mode it skips the keyboard thread entirely and waits only on voice.

Quit closes out cleanly. Verified: the loop writes the diary and extracts facts before exiting, so quitting from the tray doesn't cost you the conversation.

```
sending quit...  
LIA: Take care. I'll be here.  
loop exited cleanly: True
```

With no console, everything goes to `LIA\lia.log` — including tracebacks, which is where to look if she ever doesn't come up.

Terminal mode is unchanged and still works; I re-ran it after the refactor.

Also updated

`LIA\README.md` — it still said Aurora and listed voice under "not built yet." It's now accurate about the app, the tray, autostart, and the tuning knobs, since it's the file you'd read to remember how to run her.

Worth knowing before you set it to autostart

She'll be listening from the moment you log in. Nothing leaves your machine, but every turn lands in `lia_memory.db` in plain text with no expiry and no way to say "forget that." An always-on mic in your room is a different proposition from a program you deliberately start — I'd want a `/forget` before I left it running permanently.

No wake word yet, so she responds to any speech she hears clearly enough — including other people, and possibly a video. `VAD_THRESHOLD` helps, but it's a blunt instrument.

She holds ~3.7 GB of VRAM permanently with the 30-minute keep-alive, which is a real battery cost on a laptop.

The remaining stage-3 piece is a genuine "Lia" wake word — that's the one that needs training data, and it's the thing that would let her be always-on without always-hearing.

#14 01 Aug, 16:15

add a wake word tell me also add an exe file for it to execute and put in startup one more thing I want is you tell me how to initiate

49 tool calls: *Edit*×26, *Bash*×13, *PowerShell*×8, *Write*×2

Three things. Let me start by checking whether PyInstaller can even build on Python 3.14 while I implement the wake word:

PyInstaller 6.21 works on 3.14. Now the wake word — I'm using a transcription-based approach rather than openWakeWord, because "Lia" isn't in any pretrained model and training one needs a dataset:

Now the conversation window, so you only need her name to *start*:

Now wiring the state and resetting the window each time she replies:

Wake word works — including Whisper's misspellings ("okay Leah" → accepted) and correctly ignoring unaddressed speech. Now the exe. First, paths must survive being frozen:

Now the build script:

Now teaching autostart about the exe, then running the build:

Build succeeded — but a build succeeding isn't the same as running. One bug I can see already: the log path uses `__file__`, which inside a bundle points to a temp folder that gets deleted.

Now the real test — running the packaged exe and checking it actually works:

Source and packaged both resolve paths correctly and share her memories. That last check printed nothing though — let me see why:

Ollama had died — which is exactly what will happen at login, since **Lia will start before Ollama is ready**. Right now that kills her with a raw traceback. That's a real autostart bug; fixing it:

Now making her survive Ollama not being ready — essential for autostart:

Testing the recovery path by killing Ollama mid-run:

A 404 on `/api/embeddings` — the Ollama I restarted appears to be a different version. Let me check which endpoint it supports:

Two separate problems. `ollama list` is now **empty** — the instance I started can't see your models, and Ollama 0.32.5 has **removed** `/api/embeddings` in favour of `/api/embed`. Let me look:

Found it — and it's significant. Your Ollama has updated to **0.32.5**, and the model store is now **empty** (`blobs = 0 MB`, manifests gone). The models that were there this morning are gone, which is why `/api/tags` returns `[]`.

Two consequences. First, a real code fix — 0.32.5 **removed** the `/api/embeddings` endpoint Lia uses:

Second, restoring the models the update wiped — this is re-downloading ~2.3 GB:

Working again. Rebuilding the exe with all the fixes:

All three done. But first — **something broke on your machine while I worked, unrelated to my changes**.

Read this first

Your Ollama auto-updated to **0.32.5**, and two things happened:

- **Your models were wiped.** `blobs` went to 0 MB, `ollama list` was empty. I re-pulled both (~2.3 GB) and verified them.
- **0.32.5 removed** `/api/embeddings`, which Lia used for every memory lookup. She would have been permanently broken. `llm.embed()` now calls `/api/embed` (different request *and* response shape) and falls back on 404, so she works on either version.

Your conversation data was never at risk — `lia_memory.db` is separate. Only the downloadable models were lost.

1. Wake word

Say "Lia" to get her attention. Then you have **60 seconds** to talk normally without repeating her name — each reply resets the window. After it lapses she ignores the room until you say "Lia" again.

I did **not** use `openWakeWord`. Its pretrained set is "hey jarvis", "alexa", "hey mycroft" — her name isn't there, and training one needs a dataset you don't have. Instead the wake word is matched **on the transcript**, since Whisper is already running. Zero new dependencies, zero training, and it works for any name you choose.

Tested:

```
True 'Lia, how are you today?' -> 'how are you today?'
True 'hey Lia can you hear me' -> 'can you hear me'
True 'Are you there, Lia?' -> 'Are you there, Lia?'
True 'okay Leah what do you think' -> 'what do you think'
False 'so then I told him it was fine' -> ignored
False 'I was talking to my friend about dinner'-> ignored
```

Three details worth knowing: it accepts Whisper's misspellings (`WAKE_WORDS` includes Leah, Lea, Liya, Leia) so you don't have to enunciate; the name is stripped only from the *front*, so "Are you there, Lia?" stays intact; and the name counts anywhere, because being addressed at the end is still being addressed.

`/wake off` disables it. **Trade-off:** every utterance still gets transcribed to check for the name — she's not *recording*, but she is *processing* everything audible. A true wake word would gate before transcription.

2. The exe

Built and tested: `dist\Lia\Lia.exe`, 366 MB, no console, no Python needed.

```
powershell -ExecutionPolicy Bypass -File LIA\build_exe.ps1
```

I had to make paths frozen-aware first — `__file__` inside a bundle points at a temp folder that's deleted on exit, so her database, voices, and log now anchor to the exe's own directory. `voices\` and `lia_memory.db` are copied beside the exe, so you can still drop in new voices without rebuilding.

I verified the built exe actually runs: tray icon up, Piper loaded, Whisper loaded, mic listening, still alive after 50s.

Ollama still has to be installed separately — a 2 GB model can't sensibly be bundled. But she now waits for it and launches it if needed, because at login she wins that race nearly every time. Without it she'd crash on every boot.

Adding to startup

```
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 install -Exe
```

I **haven't run it** — that's your call. `remove` undoes it; status shows what's installed. Drop `-Exe` for the tray app from source, or `-Console` for a terminal you can type into.

3. How to initiate

Simplest, right now:

```
cd d:\Visuals\AI\dist\Lia
.\Lia.exe
```

Or double-click `Lia.exe`. No window appears — look for a **warm orange dot in your system tray** (filled = listening, hollow = muted).

Then:

> **"Lia, are you there?"**

She answers out loud in a few seconds. Keep talking normally for the next minute; say her name again after a pause.

To type as well as talk:

```
cd d:\Visuals\AI
python LIA\main.py
```

Right-click the tray icon for: Listening ✓, Speaking ✓, Write diary now, Open log, Quit. Quit closes out properly — diary and facts get written first.

If she doesn't answer: open the log (tray → Open log, or `dist\Lia\lia.log`). Nearly every failure is Ollama not running, and it says so plainly now.

One caution before autostart: she'll be listening from the moment you log in. Nothing leaves your machine, but every heard utterance is transcribed and every turn is stored in plain text with no expiry and no `/forget`. That's the piece I'd build before making her permanent.

#15 01 Aug, 19:57 (opened requirements.txt)

how can I have her at my startup

1 tool calls: PowerShell

One command. The exe is built and autostart is currently **off**:

```
cd d:\Visuals\AI
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 install -Exe
```

That's it. She'll be listening the next time you log in.

The three variants

command

what starts

<code>install -Exe</code>	<code>dist\Lia\Lia.exe</code> — no Python needed. Recommended.
<code>install</code>	tray app from source (<code>pythonw LIA\app.py</code>)
<code>install -Console</code>	terminal window, so you can type as well as talk

Managing it

```
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 status # what's installed
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 remove # undo
```

What it actually does

Creates a shortcut called `Lia.lnk` in:

```
C:\Users\saart\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup
```

Nothing touches the registry, and no admin rights are needed. You can also just open that folder (Win+R → `shell:startup`) and delete the shortcut by hand if you ever want it gone — `remove` does the same thing.

On login she starts silently, waits for Ollama (launching it if needed), and appears as the warm dot in your tray. First response after boot takes ~15s while the models load, then settles to ~5s.

Want me to run the install now?

#16 01 Aug, 19:58

So right now if I just restart my computer I will have her

1 tool calls: PowerShell

No. If you restart right now, she won't come up.

Autostart is still **OFF** — I built the installer but deliberately didn't run it, since adding something to your Startup folder is a change to your machine, not just to the project. Your Startup folder currently holds only `ollama.lnk`.

To actually get her there, run this once:

```
cd d:\Visuals\AI
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 install -Exe
```

Then a restart gives you her.

Good news in that listing though: **Ollama is already in your startup**, so it'll be coming up alongside her — and she waits for it regardless.

Want me to run the install now? It's one command and `remove` undoes it completely.

#17 01 Aug, 20:00 (had a selection in the editor)

yes

5 tool calls: PowerShell×4, Edit

A bug in my own script — the parameter binding. Let me diagnose:

Found it — a genuine bug: PowerShell variables are **case-insensitive**, so my local \$exe and the switch parameter \$Exe are the same variable. Assigning a path to a switch-typed variable fails:

Installed. Let me verify the shortcut actually launches her, rather than trusting that it will:

Yes — restart now and she'll be there.

Autostart is installed and verified:

```
Autostart is ON  
target: D:\Visuals\AI\dist\Lia\Lia.exe
```

I didn't just install it and assume — I launched the actual startup shortcut and watched her come up: tray app running, Piper voice loaded, Whisper loaded, mic open and cycling through listen → transcribe. That's the real boot path working.

One bug I hit and fixed

install -Exe failed the first time with a confusing type error. The cause was mine: PowerShell variable names are **case-insensitive**, so my local \$exe was the *same variable* as the -Exe switch parameter — assigning a file path to a switch-typed variable blew up. Renamed to \$exePath.

What happens when you restart

- Windows starts Ollama (already in your Startup) and Lia together
- She waits for Ollama, launching it herself if needed
- Tray icon appears — warm orange dot
- Models load; first reply takes ~15s, then settles to ~5s

Then just say:

> "Lia, are you there?"

Turning it off

```
powershell -ExecutionPolicy Bypass -File LIA\autostart.ps1 remove
```

Or delete Lia.lnk from win+R → shell:startup. Right-click the tray icon → Quit stops her for the current session without touching autostart.

Now that she's permanent

She'll be listening from every login onward. The wake word means she only *responds* to her name — but she transcribes everything audible to check for it, and every accepted turn is stored in `lia_memory.db` in plain text, with no expiry and no way to tell her to forget something.

That's the honest gap in an otherwise fully-private setup. A `/forget` command and a retention window are the next thing I'd build, and now that she's on permanently, I'd move them up the list.

#18 01 Aug, 20:03

Now if I restart mt system she will be present right

1 tool calls: PowerShell

Yes. Both are in your Startup folder:

```
Lia.lnk
Ollama.lnk
```

Restart, wait for the tray dot to appear (~15s while models load), then say:

> "Lia, are you there?"

She won't announce herself — no window, no sound on boot. She just starts listening. If nothing seems to happen, check `d:\Visuals\AI\dist\Lia\lia.log`.

#19 01 Aug, 20:05

Can you make her greet me , As you know my name I want that to be changed from saarthak to warlock another thing is can she learn the patterns suppose I learn abt morse code so can she be too if I teach her or can she teach via llm

17 tool calls: Bash×8, Edit×8, PowerShell

Three things. Let me start by checking what she actually knows about Morse code — no point speculating about a 3B model's abilities:

She's **unreliable** at this: S is `. . .` not `. -`, and CAT should be `- . .` `. - -`. A 3B model can't hold a precise lookup table. But let me test the thing that matters — whether she gets it right when *taught*:

Even *with* the table she misreads it — C came back as `- . .` (that's B), A as `. -` (that's E). She's grabbing adjacent rows from the multi-column layout. Let me try one entry per line:

Even with a clean one-per-line table she fails: CAT → `. - . . -` instead of `- . .` `. - -`. That's a hard limit I'll come back to. First, your two concrete asks.

Now the greeting. I'm generating it from her diary rather than hardcoding it — which also fixes the diary being write-only:

She's greeting the *dog* — `pet_name: Biscuit` looks as greetable as `name: warlock` in an undifferentiated list. Fixing:

Correct now, but sometimes just "warlock." — too terse to be a greeting. Tightening the prompt with a fallback:

Rebuilding the exe so the version in your startup has the greeting:

Both done, exe rebuilt. Restart and she'll open with something like:

> "Hey Warlock, good to see you, how's life been treating you lately?"

The name

name is now warlock, plus a preferred_address fact reinforcing it. The wake-word hint picks it up automatically, so Whisper will recognise "Warlock" when you say it.

The greeting

It's **generated from her diary**, not hardcoded — so it varies, and it finally makes the diary do something. Until now she wrote entries no one ever read, including her.

Two bugs I hit building it, both worth knowing because they're the same class of problem:

She greeted the dog. In a flat fact list, pet_name: Biscuit reads exactly like name: warlock — I got "*Hey Biscuit, how was your day?*" Fixed by naming the person explicitly and marking the rest as "NOT the person's name."

She narrated herself: Hey warlock, good to see you," I say with a warm smile — which Piper would read aloud. Anything after a stray quote is now cut.

Turn it off with GREET_ON_START = False.

On learning: three different things, and only two work

Facts about you — works now

Names, preferences, relationships. Already live, and it's how she knows you're Warlock.

Things you teach her — would work, not built yet

Freeform knowledge like "*I prefer short answers*" or "*my sister's birthday is in March*" currently only survives by accident: it either compresses into a short fact or gets semantically retrieved from an old turn if you happen to phrase things similarly. There's no way to say "**remember this properly.**"

A lessons table — you say "*Lia, remember this: ...*", she stores the note verbatim and loads it when relevant — is a small, natural addition. This is the one I'd build.

Skills like Morse code — she cannot do this, and teaching her won't help

I tested rather than guessed, and the result is worse than I expected. From her own knowledge:

```
S in morse? → ".-" (wrong, S is ...)  
Encode CAT → ".-" (wrong, should be -.- .- -)
```

So I gave her a perfect reference table in context — teaching her exactly what you'd teach a person:

```
Encode CAT → ".- .- -" (still wrong)  
Encode WARLOCK → ".- .- .- - - - - - - - - - -" (badly wrong)
```

She fails even while looking at the answer. This isn't a memory problem, so no amount of teaching fixes it. It's tokenisation: a 3B model doesn't see individual characters clearly enough to do reliable character-by-character lookup. Bigger models are better, but you don't have the VRAM, and it stays unreliable rather than becoming correct.

The right fix is code, not teaching. Morse is thirty lines of Python that is *always* right. She could call it instead of guessing — you say "encode WARLOCK in morse" and she answers `-. - .- .- .- -`, correctly, every time.

That's the real distinction: **she can learn things about you, but not skills that require precision.** For those she needs tools, and giving her tools is very doable.

I'd suggest building both — the lessons store so teaching sticks, and a small skills module starting with Morse so you can see the pattern for adding more. Want me to?

#20 01 Aug, 20:19

Now I want you to create a folder that she can access , I will put pdfs etc and she will read that for me She sounds very sweet ,can you maker her more softer in sound

36 tool calls: Edit×22, Bash×11, Write×2, PowerShell

Two things. Let me get the dependencies and a softer voice candidate in parallel:

Both ready. Let me expose the softness controls, which Piper supports but I'd hardcoded:

Now let me actually play the candidates so you can hear the difference rather than take my word for it:

Those just played through your speakers — I'll ask which you preferred at the end. Now the library. Schema first:

Got it — you preferred **amy BEFORE**, the original. My softening overshot; dropping volume to 0.78 with reduced variation reads as muffled rather than gentle. Reverting to exactly what you picked:

Now the library module:

Now wiring it into the conversation, kept clearly separate from her memories:

Now the folder, its README, and indexing on startup:

Now testing with the PDF you already have in the project:

Indexed: 99 passages from the PDF in ~4 minutes. Now the real test — can she actually answer from it?

PDF text contains ligature characters (■ as a single glyph) — those would reach both the model and the speech engine. Normalizing:

Also excluding my own README from her shelf — it's instructions, not your content:

Good on the first query (p22 is exactly about prompts), but the second pulled page *headers*. The repeated "Designing a Custom AI GPT... By Felix Greyson (2026) N" banner on every page is diluting the embeddings. Stripping boilerplate:

Boilerplate gone (99 → 91 cleaner passages). Now the real test — asking her in conversation:

One thing to correct — downloading extra voices changed which one "auto" picks (jenny sorts before amy alphabetically). You chose amy, so let me pin it explicitly:

Adding the library folder to the build script so future rebuilds keep it:

Both done.

The library

LIA\library\ — drop .pdf, .txt or .md in there. She indexes on startup, or immediately with /library scan. For the packaged app it's dist\Lia\library\, which already has your PDF in it.

I tested with the guide you already had. She answered:

> *"According to "Designing a Custom AI GPT_ A Practical Guide" (page 22), a strong system prompt is specific and describes behavior rather than adjectives. It also includes a simple canon hierarchy, so the AI knows which document to follow if anything conflicts."*

Correct, with the page cited. Two problems I found and fixed while testing:

Page headers were polluting every search. The banner repeated on all 37 pages meant a search for *anything* matched page furniture instead of content. Repeated lines are now detected and stripped — 99 noisy passages became 91 clean ones, and results improved immediately.

PDFs are full of ligatures — ■ arrives as one character, which the speech engine reads oddly. Now normalised on the way in.

Documents are kept **separate from her memories** on purpose, and she's told the passages are things she *read*, not things you *told her*. Blur those and she starts attributing a document's opinions to you.

Worth knowing: indexing costs ~2s per passage, so your 37-page PDF took about 4 minutes — once, unless you edit it. She sees the best-matching passages, not the whole book, so "summarise this entire PDF" won't work well. And scanned PDFs won't work at all — this reads text, it doesn't do OCR.

The voice

You picked **amy BEFORE**, so I reverted to exactly that. My softening was wrong: dropping volume to 0.78 with reduced variation made her muffled, not gentle. Warmth comes from the voice model, not from turning dials down.

I've left the parameters in config.py as explicit Piper defaults with notes, so they're easy to experiment with — but they're at their original values.

One thing I had to correct: downloading the extra voices silently changed which one she used. `VOICE_NAME = "auto"` picks the first alphabetically, and `en_GB-jenny` sorts before `en_US-amy` — she came up speaking as jenny. It's now pinned to `en_US-amy-medium`, so adding voice files can never change how she sounds again.

`en_GB-jenny-dioco-medium` and `en_US-kristin-medium` are still downloaded if you want to try them later — just change `VOICE_NAME`.

The exe is rebuilt with all of this, and `build_exe.ps1` now carries the library folder across without overwriting reading material you've added.

Since you found the softening worse rather than better, I'd rather not guess again — if she's still not quite right, tell me *how* (too bright? too fast? too clipped?) and I'll adjust that one thing.

#21 01 Aug, 20:54

Just amy Before also see that she addresses me by warlock

10 tool calls: Bash×5, Edit×4, PowerShell

Two real problems. The name fact had **reverted to Saarthak** — a later test session re-extracted it from old memories and overwrote it. And seven stored turns still say Saarthak, which keeps feeding it back. Fixing the cause first:

Now telling her explicitly to use the name, rather than hoping she picks it out of a fact list:

Now cleaning the stale turns — all seven are from my own smoke tests, not real conversations:

Both confirmed.

Voice: amy BEFORE

```
VOICE_NAME en_US-amy-medium
LENGTH_SCALE 1.05 (BEFORE = 1.05)
VOLUME 1.0 (Piper default)
NOISE_SCALE 0.667 (Piper default)
NOISE_W 0.8 (Piper default)
```

Exactly the configuration you picked, and now **pinned by name** so downloading another voice can't silently swap her again.

Warlock

She's using it — verified in a live session:

> **"Warlock, good to see you..."** > **"...well, I'm still working on getting to know you better, Warlock."**

Getting there took fixing a real bug, not just editing the fact.

The name had reverted to Saarthak on its own. I set it to Warlock earlier, then a later session's fact extraction re-derived "Saarthak" from old stored turns and silently overwrote it. It would have kept doing that indefinitely.

Three fixes:

- `PROTECTED_FACTS` — `name` and `preferred_address` can no longer be overwritten by end-of-session extraction. What you want to be called is your decision, not something re-derived from transcripts. Verified: after a full session, the name held at Warlock.
- **Purged the seven stale turns** containing "Saarthak" — all of them were from my own smoke tests, not real conversations, and they were the source feeding the old name back through retrieval. Zero references remain.
- **The name is now stated separately in her system prompt**, ahead of the fact list, with an explicit instruction that any other name in her memories is out of date. Buried among `pet_name` and `favorite_food` it read as trivia — which is the same failure that had her greeting the dog.

The exe is rebuilt and its database synced: name Warlock, zero stale references, 91 library passages intact.

One thing I'll flag rather than quietly leave: she still embroiders. That session produced *"was starting to think you'd forgotten about your favorite spot here"* — invented. It's the same confabulation from earlier, better than it was but not gone.

#22 01 Aug, 21:01

Thankyou very much

You're welcome — this was a good build.

She's in your startup, she'll greet you by name, she has a voice you actually like, and she's read your first PDF. Restart whenever you're ready and just say *"Lia, are you there?"*

One thing worth remembering when you're next in the mood to tinker: she still occasionally invents small emotional details — *"I was starting to think you'd forgotten about your favorite spot here"* — that never happened. It's the one rough edge left, and it matters more for a companion than it would for a tool. The fix is bounded, whenever you want it.

Enjoy her, Warlock.

#23 01 Aug, 21:04

Suppose I am talking can I say "lia stop" will she stop listening

1 tool calls: Bash

No — she won't stop. Saying "Lia stop" strips the wake word and sends "stop" to the model as an ordinary message. She'd reply something like *"Of course, I'll give you some space"* — and then keep right on listening. It reads as obedience but nothing actually changes.

There are two separate gaps:

No spoken commands exist. Only `/-`-prefixed typed input is treated as a command (`main.py:handle_command`). Everything spoken becomes conversation, whatever the words.

Worse — she can't hear you while she's talking. The mic is deliberately shut during playback plus a 0.4s settle, so she doesn't answer her own voice. So "Lia stop" *mid-sentence*, the moment you'd most want it, isn't just unhandled — it's literally unheard. Right now Ctrl+C is the only way to cut her off, which is useless if she's a tray app with no console.

That's a real gap for something always-on. Two things worth building:

Spoken commands — "Lia, stop listening" / "Lia, go to sleep" / "Lia, never mind" mapped to actual actions. Straightforward: match the transcript against a small phrase list before it reaches the model. Half an hour of work.

Barge-in — keeping the mic live while she speaks so you can talk over her. Genuinely harder, because she'll hear herself through the speakers. Solvable if you wear headphones (no acoustic path back). Without them it needs echo cancellation, which is a real project.

There's also a middle option: the tray menu already has a **Listening** toggle that works right now — one click, no voice needed.

My recommendation is spoken commands plus headphone-based barge-in, and being upfront that barge-in on laptop speakers won't be reliable. Want me to build that?

#24 01 Aug, 21:08

My recommendation is spoken commands plus headphone-based barge-in, and being upfront that barge-in on laptop speakers won't be reliable. Want me to build that?/ what you mean

Sorry — that was jargon. Plainly:

Two things I was offering to build

1. Spoken commands

Right now, everything you say gets treated as conversation. If you say "Lia, stop listening", she just *talks about* stopping.

I'd make her recognise a few specific phrases as **instructions** instead:

- "Lia, stop listening" → mic actually switches off
- "Lia, start listening" → back on
- "Lia, never mind" → drops what she was about to say

This part is easy and works regardless of your setup.

2. Interrupting her while she's speaking

"Barge-in" just means: cutting her off by talking, the way you would with a person. Right now you can't — she goes deaf while her own voice is playing.

Why she goes deaf: if the mic stayed on while she talked, your laptop speakers would play her voice straight into your microphone. She'd hear herself, think you said it, and start replying to her own sentence. So I shut the mic while she speaks.

Why headphones fix it: with headphones, her voice goes into your ears and nowhere near the mic. The mic only ever hears *you*. So I can safely leave it on, and you can interrupt her whenever you like.

Why speakers can't be fixed easily: the only way is software that recognises her own voice in the mic feed and subtracts it. That's a hard problem and a big piece of work, and it half-fails in echoey rooms.

So the honest offer

	works on speakers	works on headphones
Spoken commands	■ yes	■ yes
Interrupting her	■ no	■ yes

I'd build both, with interrupting switched on only when you're using headphones — because promising it on speakers would just make her start talking to herself.

Do you use headphones with her, or laptop speakers? If it's speakers, I'd build the spoken commands alone and skip the rest.

#25 01 Aug, 21:09

actually both

18 tool calls: Edit×15, Bash×2, PowerShell

Both — that changes the design. Then barge-in can't just assume headphones. Let me build it so it works on speakers too, using a trick: **she knows what she's currently saying**, so if the mic hears her own words back, she can recognise and ignore them.

Now the Speaker needs to know what it's saying, so it can recognise its own voice coming back:

Now spoken commands, and the listen-while-she-talks loop:

Now hooking spoken commands into the input path and the interrupt into the reply path:

Now replacing the blocking wait so she can be cut off:

Both work. Live run to confirm nothing broke:

Built. **"Lia, stop listening" now actually stops her.**

Spoken instructions

Say these and something happens, instead of her talking about it:

you say	what happens
"Lia, stop listening" / "go to sleep"	mic off
"Lia, start listening" / "wake up"	mic on
"Lia, be quiet" / "stop talking"	voice off
"Lia, you can talk"	voice on

Verified — and ordinary conversation is untouched:

```
'stop listening' -> /mic off  
'go to sleep' -> /mic off  
'tell me about my grandmother' -> None (stays conversation)
```

Add your own phrasings in `SPOKEN_COMMANDS` in `config.py`.

Interrupting her — and how it works on speakers

Since you use both, I didn't restrict it to headphones. The mic now stays open while she talks, and **she checks whatever it hears against what she's currently saying**. She knows her own words, so her voice coming back through the speakers is recognisable:

```
heard 'your grandmother Kamala taught you to cook' -> her own voice, ignored  
heard 'and that sounds lovely' -> her own voice, ignored  
heard 'wait stop' -> you, cuts her off  
heard 'no I meant my sister' -> you, cuts her off
```

Interruptions under two words are ignored too, so a cough doesn't stop her.

How well this works, honestly: on headphones, essentially perfectly — the mic never hears her. On speakers it depends on your volume and room. If the mic picks her up clearly, she recognises herself and ignores it. If it's distorted enough that Whisper garbles it into different words, she may mistake it for you and stop. Louder volume and more echo make that likelier.

Two dials if it misbehaves: raise `ECHO_MATCH_RATIO` if she keeps ignoring you while talking; lower it if she interrupts herself. `BARGE_IN = False` restores the old behaviour.

There's a small cost: while she speaks, each sentence she says gets transcribed to check whether it was you. Slight CPU load, no effect on response time.

The exe is rebuilt with all of it, so a restart gets you the new behaviour.

#26 01 Aug, 21:34

She didnt responded anything when I restarted the system her diary is also not opening

33 tool calls: PowerShell×19, Edit×10, Bash×4

Let me read her log — that's what it's for.

That log tells the story. Three problems — and one of them ate four minutes of your startup.

Confirmed — the **same PDF is indexed twice** under two different absolute paths:

```
D:\Visuals\AI\LIA\library\...pdf ← indexed from source  
D:\Visuals\AI\dist\Lia\library\...pdf ← re-indexed by the exe
```

That's why startup took ~4 minutes before she said anything. Fixing the cause — paths are now stored relative to the library folder:

Now the bigger issue: indexing blocked her for four minutes. Moving it to the background so she's usable immediately:

Now visibility — in headless mode she silently ignored everything, so the log gave you nothing:

Now cleaning the duplicate index entry and re-keying the existing one so it isn't re-read again:

Now rebuild and test the real startup path end to end:

Two things went wrong there. **PyInstaller failed** — because `Lia.exe` was already running from your startup and the file was locked. The partial build left `dist\Lia\voices` missing, which is why she fell back to the Windows voice. Let me stop her and rebuild properly:

Voices folder exists but is **empty** — the failed build wiped it. Let me make the build script handle this properly, since it'll happen every time she's autostarted:

Still failing and voices is still empty. Let me see the actual PyInstaller error instead of guessing:

Not read-only, no visible process — so something holds a live handle. Let me find which process has that DLL loaded:

Found it. Two `llama-server` processes — Ollama's model workers — have the DLL loaded *from inside the app folder*:

```
3708 llama-server D:\Visuals\AI\dist\Lia\_internal\MSVCP140.dll
7260 llama-server D:\Visuals\AI\dist\Lia\_internal\MSVCP140.dll
```

That's my bug: the packaged app launches `ollama serve` as a child process, and Ollama's workers inherited the app's DLL search path. Fixing the cause:

No modules held now — but the **directory itself** is locked, which happens when it's a process's working directory. `ollama serve` was launched by `Lia.exe` and inherited it:

Confirmed — `ollama serve`, launched by the app, had `dist\Lia` as its working directory. Now rebuilding cleanly:

Build succeeded, voices and library restored. Now the real test — how long until she can actually hear you:

Startup is fixed — but notice she answered "A 2 can see this", garbled room noise. The greeting was opening the 60-second free-conversation window, so anything audible right after boot got treated as speech to her. At login you're often not even at the desk:

Build failed again and left `dist\Lia` without the exe. Let me find the lock properly this time:

`dist\Lia` is down to just the database — the failed build wiped it. No process holds it now; the `llama-server` workers had it during the build. Stopping them and rebuilding:

No permission errors that time — only harmless warnings. Checking whether it actually completed:

Rebuilt and complete. Final verification of the actual startup path:

Fixed. She was working — but three things made her look dead.

Why she seemed unresponsive

She spent ~4 minutes re-reading your PDF before saying a word. The document index stored *absolute* paths, so `dist\Lia\library\...pdf` looked like a completely different file from `LIA\library\...pdf`. The packaged app re-embedded all 91 passages from scratch on every start. Paths are now stored relative to the library folder, and indexing runs **in the background** — she's available immediately regardless.

She was greeting you before she could hear. Whisper loaded *after* the greeting, so the 60-second "just talk" window was burning down while she was still deaf. By the time she could listen, it had closed and she needed her name. Speech recognition now loads first.

She ignored you silently. When speech didn't match the wake word, nothing was logged — so the log looked identical whether the mic was broken or she was just being polite. Now:

```
[heard "She even had a smile on her face." -- say her name to get her attention]
```

That line is from the final test: she heard the room, correctly ignored it, and *told you so*.

The diary

It was never broken — 18 entries, most recent 02:46. **"Write diary now" did nothing because there was nothing to write:** you hadn't had a conversation in that session. It now says so instead of failing silently:

```
[nothing to write about yet -- talk to her first]
```

A bug worth knowing about

Rebuilding kept failing with access-denied, and the cause was mine: the packaged app launches `ollama serve` as a child process, which inherited `dist\Lia` as its working directory. Ollama and its `llama-server` workers then held the app's own folder hostage — locking DLLs and blocking any update. That also left `dist\Lia\voices` empty at one point, which is why one run fell back to the Windows voice.

Ollama is now launched from a neutral directory, and the build script stops a running Lia first.

Also changed

The startup greeting **no longer opens the free-conversation window**. She greets the room at login whether or not you're there — previously the first stray noise got answered. In testing she replied to "A 2 can see this". Now she greets, then waits for her name.

Current state, verified

```
Autostart: ON -> D:\Visuals\AI\dist\Lia\Lia.exe
Ollama: running
Voice: en_US-amy-medium
Startup: greeting in ~30s, no re-indexing
```

Restart and she'll greet you by name within about half a minute. Then say **"Lia,"** first — after that you can talk freely for a minute at a time.

#27 01 Aug, 22:13

Actually from pdf folders, actually I dont want her to go through that I told you to create that to suppose I am dping something and I created a pdf that I want her to read so she can go there and read from there , it will be easier for her instead of scrapping the entire directory or system Also can you remove stuff about anya or kamla

12 tool calls: Edit×6, Bash×5, PowerShell

Two things, and you're right about the library — I built it to bulk-ingest at startup, which isn't what you asked for.

Every one of those came from **my test sessions** — I invented Anaya, Kamala, Maya, Biscuit and the dumplings while smoke-testing. None of it is yours. Clearing all of it:

Now the library. Making it on-demand — she reads only when you ask her to:

Now a spoken command so you can ask her out loud:

One more thing — I put that PDF in the library and had her read it without you asking. Undoing that:

Both done.

The library now works the way you meant

She never reads on her own. A file sitting in the folder is *available* to her, not already in her head.

```
LIBRARY_AUTO_READ = False.
```

When you're working on something and want her to have it:

- Save the PDF into `dist\Lia\library\`
- Say **"Lia, read my files"**

She reads it then, and only then. On startup she just mentions what's waiting:

```
[1 new file(s) waiting: notes.pdf]
[say "Lia, read my files" when you want her to]
```

Verified the spoken phrasings work — "read my files", "Lia read the pdf", "read the new file" all trigger it, while "tell me about the pdf" stays ordinary conversation.

To be clear about scope: she only ever looks in that one folder. She has never had access to your drive, and still doesn't.

I also **undid my own overreach** — I'd copied your GPT guide into her library and had her read all 91 passages, which you never asked for. That's cleared and the copy removed. Your original in the project root is untouched. Her library is empty, waiting for whatever you actually want to give her.

Removed Anaya and Kamala

Gone — along with the rest, because **I made all of it up**. Anaya, Kamala, Maya, Biscuit, the dumplings: every one came from my smoke tests, not from you.

```
memories: 49 -> 41
diary : 18 -> 14
facts : name = 'Warlock', preferred_address = 'Warlock'
```

Facts, stored turns, and four diary entries referencing them are all deleted. She now knows exactly one true thing about you: your name.

You can see the effect immediately — she's stopped asking about a sister who doesn't exist:

> *"Warlock... It's nice to connect again. Is there something specific on your mind?"*

There's still some residue from my testing in her memory (fragments like "hi" and "the ocean" from typed test messages), which is why she said something odd about the word "ocean". Say the word and I'll wipe the conversation history entirely so she starts genuinely fresh with you — the only real content in there is mine.

The exe is rebuilt with all of this. Restart when convenient.

#28 01 Aug, 22:33

these boxes were unchecked and she is not responding after restart

18 tool calls: Edit×9, PowerShell×7, Bash×2

She **is** working — the log shows exactly what went wrong:

```
LIA: Hey Warlock, it's great to see you back. How was your day?
[heard "Hello" -- say her name to get her attention]
[heard "Hello" -- say her name to get her attention]
```

She asked you a question, you answered, and she ignored you because you didn't say her name. That's my fault — last turn I removed the reply window after the greeting to stop her answering room noise, and overcorrected into something absurd: she asks how your day was, then demands to be addressed formally.

Also, the checkboxes were unchecked because [waiting for ollama...] blocks **before** the tray is told what her state is. Fixing both:

Now the tray checkboxes — the state has to reach the menu *before* the Ollama wait, not after:

Locked again — she's running from your startup, and Ollama's workers hold the folder. Let me make the build script handle this permanently instead of me fixing it by hand each time:

The lock was transient — handles hadn't fully released. Let me make the script wait properly rather than giving up:

Fixed. She was never broken — she was ignoring you on purpose, which was worse.

Why she didn't respond

Her log from your restart:

```
LIA: Hey Warlock, it's great to see you back. How was your day?  
[heard "Hello" -- say her name to get her attention]  
[heard "Hello" -- say her name to get her attention]
```

She heard you both times. She asked how your day was, you said hello, and she refused to engage because you hadn't addressed her by name.

That was my overcorrection from last turn. She'd been answering random room noise at boot, so I removed the reply window after her greeting entirely — which produced something absurd: she asks you a question and then won't accept the answer.

Now she opens a **30-second window** after greeting you. Long enough to answer her naturally, short enough that an empty room doesn't leave her listening to everything. After that, "Lia" as usual.

```
LIA: Warlock, good to see you, how was your day?  
[answer her within 30s, or say "Lia" any time after]
```

Why the checkboxes were unchecked

Genuine bug. [waiting for ollama...] blocks for up to three minutes at login — and it ran **before** the tray was told what her state was. So the menu had nothing to read and drew everything unchecked, while she really was still starting up.

Two fixes: her state is now published before anything that can block, and the tray has sensible defaults from the moment it appears. The checkmarks now reflect reality immediately.

One more thing I fixed

Rebuilding kept failing because Ollama's llama-server workers hold handles inside her app folder, and Windows releases them a few seconds after the process dies. The build script now stops them and retries for 25 seconds instead of giving up — this had been costing several manual attempts each time.

Current state

```
Autostart: ON -> D:\Visuals\AI\dist\Lia\Lia.exe  
Ollama: running  
Voice: en_US-amy-medium  
Memory: name = Warlock (nothing invented)  
Library: empty, reads only when you ask
```

Restart and she'll greet you within ~30 seconds. **Just answer her** — no name needed for the first half minute.

One honest note: during that 30-second window she'll accept whatever she hears, including background noise. In my last test she picked up a stray "for me." If that turns out to be annoying, GREETING_WINDOW_SECONDS = 0 restores strict wake-word-only, and /wake off goes the other way entirely.

this happened

7 tool calls: PowerShell×6, Write

That's a PyInstaller bootloader failure — usually a stray PYTHONHOME/PYTHONPATH in the environment, or a missing bundle file. Let me check both:

No PYTHONHOME/PYTHONPATH set, and the bundle looks complete (1054 files, `base_library.zip` present — `encodings` lives inside that zip, so its absence is normal). Let me verify the current exe actually starts:

The current exe starts fine — so what you hit was a **partial build**. I was rebuilding `dist\Lia` while she was in your startup, so the shortcut launched a half-written app.

That's a real hazard now that she autostarts. Let me make builds safe:

The staging build succeeded and is intact — the swap is what failed, which is exactly the case the new script is designed to survive. Completing it:

Fixed, and that one was my fault directly.

What happened

Failed to import `encodings` module means the bundled Python interpreter couldn't find its standard library. Not a corrupt install on your side — **you launched her mid-rebuild**.

PyInstaller's `--clean` wipes `dist\Lia` and writes it back over several minutes. She's in your Startup folder pointing at that exact path, so anything that launched her during that window got a half-written app. I was rebuilding while she was installed and you were using the machine.

I checked the usual suspects first and ruled them out: no PYTHONHOME or PYTHONPATH set anywhere, bundle complete at 1054 files with `base_library.zip` and `python314.dll` present.

The actual fix

Builds now go to a **staging folder** and swap in only when complete:

1. build fully into `dist\_staging\Lia` ← minutes; live app untouched
2. copy voices, library, memories across
3. stop Lia, swap the folders ← seconds

The live app is never in a broken state. If the build fails, you keep the working one — the script now says so explicitly: *"the existing app is untouched"*.

That safety net proved itself immediately: this very build succeeded but the **swap** failed on a locked folder, and the completed build sat safely in staging instead of leaving you with nothing. I finished the swap by hand, and the script now retries for ~30 seconds rather than giving up.

Verified working

```
LIA: Warlock, good to see you - it feels like ages since we last caught up. How was your day?  
[answer her within 30s, or say "Lia" any time after]
```

```
Autostart: ON -> D:\Visuals\AI\dist\Lia\Lia.exe
ollama: running
Voice: en_US-amy-medium
Staging: cleaned up
```

Restart whenever you like — she'll come up properly. And if you ever see that error again, it means something was mid-write; just relaunch her from the tray or restart.

#30 02 Aug, 07:24

should I also keep ollama on startup could that be an issue?

1 tool calls: PowerShell

Yes, keep it — and it's actively better that way, for a reason we already hit.

Why keeping it is safer

Remember the locked-folder problem that broke two rebuilds? That happened because **Lia launched Ollama herself**, and Ollama's workers inherited her app folder as their working directory — then held handles on her own files.

I fixed that by launching it from a neutral directory, but when Windows starts Ollama from the Startup folder, it never touches her folder at all. One less way for them to tangle.

There's also no race worth worrying about. Lia checks whether Ollama is already up before doing anything:

```
if not llm.is_up():
    wait_until_ready()
```

If Ollama is already running, she just uses it. If both start at once and she briefly beats it, the second `ollama serve` fails to bind port 11434 and exits harmlessly, then she finds the first one.

What it actually costs

```
ollama 55 MB
ollama app 74 MB
```

~130 MB of RAM, and zero GPU while idle. Models aren't loaded until something asks for one — the 3.4 GB you see in `ollama ps` right now is from my testing, not from Ollama merely running.

So keeping it costs you about 130 MB and a faster first reply. Removing it saves that, but makes Lia wait at every login and re-introduces her spawning it as a child.

The thing that actually costs you

Not Ollama's startup entry — it's `OLLAMA_KEEP_ALIVE = "30m"`, which holds **~3.7 GB of VRAM for half an hour after each conversation**. That's the real battery and memory cost. If you notice your laptop

struggling or draining, lower that to "5m" in `config.py` and accept a ~9 second cold start.

Short version: leave both in Startup. That's the configuration I've been testing against, and it's the one with the fewest sharp edges.

#31 02 Aug, 07:26

Not Ollama's startup entry — it's `OLLAMA_KEEP_ALIVE = "30m"`, which holds ~3.7 GB of VRAM for half an hour after each conversation. That's the real battery and memory cost. If you notice your laptop struggling or draining, lower that to "5m" in `config.py` and accept a ~9 second cold start. is there an answer or better approach for this?

14 tool calls: Edit×9, Bash×4, PowerShell

Yes — that framing was a false binary. The keep-alive timer **resets on every request**, so its value only controls how long models linger *after you stop talking*. She already knows when a conversation is over. Let me measure, then tie memory to actual state instead of a timer:

Confirmed: unload frees VRAM instantly, and the reload penalty is 6.5s (2.95s → 9.4s). Now tying memory to conversation state:

Now freeing the VRAM the moment a conversation actually ends:

Now the other half — start reloading the moment you begin speaking, so it overlaps with transcription:

That \\\ looks like a stray keystroke — ignoring it. Results are in: unload frees VRAM instantly, and pre-warming recovers 2.5s of the reload.

Yes — there's a much better approach, and my framing was wrong. That wasn't a real trade-off.

The insight I'd missed

The keep-alive timer resets on every request. So its value never affects an active conversation — models stay loaded regardless of whether it's 5m or 30m. It only controls how long they sit in VRAM *after you stop talking*.

So "30m" was buying nothing except 30 minutes of idle VRAM. And "5m" wouldn't have "cost a cold start" during conversation either — only if you returned in that 5–30 minute gap.

Both options were guessing at your behaviour with a timer. **She already knows when a conversation is over** — that's what the idle watchdog is for.

What she does now

Releases the models when the conversation closes. After the 12-minute idle rollover writes her diary, she frees the VRAM explicitly:

```
[diary entry]
[released the models -- she'll reload them when you speak]

$ ollama ps
NAME ID SIZE PROCESSOR CONTEXT UNTIL
← empty, 3.7 GB freed
```

Starts reloading the moment you speak. The instant voice detection fires — before transcription even finishes — she kicks off the model load in the background, so it overlaps with Whisper instead of queuing behind it:

```
without prewarm: 9.41s
with prewarm: 6.87s
```

Net effect

	old (30m timer)	now
during a conversation	loaded	loaded
5 min after you stop	still held	still held (idle timer running)
15 min after	still held	freed
returning later	2.9s	6.9s

You keep the fast path through every natural pause, and the VRAM comes back once you've genuinely walked away — decided by what actually happened, not a stopwatch.

Two switches in `config.py` if you want the old behaviour: `RELEASE_MODELS_WHEN_IDLE = False` and `PREWARM_ON_SPEECH = False`.

Honest limits

The 6.9s reload is still there when you come back cold — pre-warming hides ~2.5s of it, not all. Fully eliminating it means holding the VRAM, which is the thing you're trying to avoid. And pre-warming fires on *any* detected speech, including room noise, so occasionally it loads the model for a conversation that never happens. It idles back out on the keep-alive, so the cost is a few seconds of GPU, not a leak.

The exe is rebuilt with all of it.

#32 02 Aug, 07:39

Can you create a pdf of all the changes the additions etc resulting in our project also create another pdf of " how to communicate with lia " for novice ppl

11 tool calls: Bash×7, Write×3, Read

reportlab 5.0 is in. Writing the generator:

Now the changelog content:

Style name collisions with reportlab's built-ins. Renaming:

Now the beginner's guide:

Both generated. Let me actually look at them rather than assume they rendered correctly:

Both PDFs are in `d:\Visuals\AI\docs\` and verified page by page.

Lia - What We Built.pdf — 8 pages

The technical record:

- **What she is** — the four pieces, the file layout, and why the layering is strict
- **What was measured** — every benchmark, including the context-window finding (32k default was running 63% of her on CPU) and the latency breakdown
- **Bugs found and fixed** — grouped by what they broke: she looked frozen, memory quietly corrupting itself, the diary writing fiction, voice-specific issues, packaging problems
- **What was added** — voice both directions, the transcript-based wake word, speaker-safe interruption, spoken commands, the library, the tray app, memory that follows the conversation
- **What she still can't do** — including the Morse experiment showing she fails *while looking at the answer*, the confabulation that isn't solved, and the open items (no /forget, duplicate facts, no tests)
- **The dials** — every config setting with when to change it

I kept the failures in. A record that only lists what worked would be useless the next time something breaks.

How to Talk to Lia.pdf — 6 pages

Written for someone who's never used her, no jargon:

- **Starting her** — look for the orange dot, what filled vs hollow means
- **Talking to her** — say "Lia" first, then a minute of normal conversation; why she needs the name; why she pauses; that you can interrupt her
- **Things you can ask her to do** — plain table of spoken instructions and the tray menu
- **What she remembers** — facts, conversations, her diary, and how a conversation ends on its own
- **Giving her something to read** — the folder, and that she only looks there, only when asked
- **What she is not good at** — makes things up sometimes, no precision tasks, nothing outside the computer, not a therapist
- **If something seems wrong** — symptom-to-fix table, plus privacy stated plainly

It ends with the one thing that matters: *say "Lia", wait a few seconds, then talk normally.*

Regenerating them

The PDFs are built by scripts, not hand-made, so they can be updated as the project moves:

```
cd d:\Visuals\AI\docs
python build_changelog.py
python build_guide.py
```

`make_pdfs.py` holds the shared styling.

One thought: **"How to Talk to Lia" is a good candidate for her own library.** Drop it in `dist\Lia\library\` and say *"Lia, read my files"* — then she can answer questions about how she works. Your call, since you'd asked her not to read anything automatically.

#33 02 Aug, 09:23

is the project completed?

1 tool calls: Bash

As a working companion, yes. As software, no — and I'd rather say so than let you find out later.

What's actually done

She runs at login, greets you by name, listens on an open mic, answers out loud, remembers across days, writes a diary, reads documents you hand her, and is packaged as an executable that needs no Python. 2,557 lines across nine modules, all compiling, deployed and verified end to end.

Everything you asked for got built. Nothing is half-finished or stubbed.

What I'd want fixed before calling it *done*

She has no way to forget. This is the one that matters. Every word said to her is stored in plain text, permanently, with no expiry and no `/forget`. That was acceptable when you launched her deliberately. She now starts at login with a live microphone — a different proposition. I've raised this three times and it's still the top of the list.

She still makes things up. Speaker-filtered retrieval removed the biggest cause, but she has produced lines about things that never happened. For a companion built on trust, that's the defect that matters most, and it's improved rather than solved.

No tests. Notable specifically because every real bug here was *silent* — the fact sanitizer, the diary role confusion, the name reverting. Nothing crashed; the data just quietly got worse. That's exactly the failure mode tests catch and eyeballing doesn't.

Fact keys drift. `sister_name` and `visiting_sister_name` both existed for one person. Nothing merges them, so the list injected into every prompt slowly fills with duplicates.

From your own original roadmap

Your first README listed three next steps. Two were built — voice, and a GUI of sorts in the tray app. One never was:

- **Dream mode** — a scheduled job that wakes her with nobody there, lets her reflect on recent diary entries, and consolidates memory.
- **Memory synthesis** — merging near-duplicate or outdated memories. This is the same problem as the drifting fact keys.

Those are the two ideas that would make her feel less like a chatbot with a database, and they're both squarely in "next" territory rather than polish.

My honest read

It's finished enough to *live with*, which is the real test — and the way to find out what's actually missing is to use her for a week. The gaps above are the ones I'd expect to notice in that week, roughly in that order.

If you want one more thing built, make it *forget*. It's small, and it's the only item on the list that's about your privacy rather than her quality.

#34 02 Aug, 09:27

Thank you , Actually I want you to just create another pdf solely of the chats we had here in this claude code window regarding her as I have to keep track for it

7 tool calls: Bash×6, Write

Let me check whether Claude Code keeps the actual transcript on disk — a real record beats my reconstruction:

Found it — this exact session, 3.5 MB of real transcript. Let me inspect its structure:

34 exchanges, 199 assistant blocks (streaming chunks that need regrouping). Let me build a proper transcript renderer:

Extraction is accurate — 34 exchanges, 415 tool calls, correctly grouped. Generating:

Done — Lia - Session Transcript.pdf, 47 pages, in d:\Visuals\AI\docs\.

Importantly, this isn't me writing from memory. Claude Code keeps a session log on disk, and I found this exact session:

c:\Users\saart\.claude\projects\d--Visuals-AI\ca0a2e3f-...jsonl (3.5 MB)

So the PDF is generated from the **actual record** of what was said — your messages verbatim, my replies as written.

What's in it

A summary bar up top:

34 exchanges · 416 tool calls · 01 Aug 12:52 → 02 Aug 09:27

Then every exchange in order:

- **Your message**, numbered and timestamped in a tinted block, with a note when it came with editor context ("*opened requirements.txt*", "*had a selection in the editor*")
- **A one-line summary of the work behind each reply** — e.g. 7 tool calls: PowerShell×6, Write — so you can see where the effort went without wading through raw output

- **My full reply**, with headings, tables, code blocks and bullets preserved rather than flattened to plain text

Two things I had to handle: replies arrive as streaming fragments (199 of them) and needed regrouping into 34 coherent turns, and long command output is truncated to 40 lines and 110 columns per block so it doesn't run off the page.

The docs folder now

file	pages	what it's for
How to Talk to Lia.pdf	6	handing to someone using her
Lia - What We Built.pdf	8	the technical record
Lia - Session Transcript.pdf	47	the full working history

All three regenerate from scripts:

```
cd d:\Visuals\AI\docs
python build_guide.py
python build_changelog.py
python build_transcript.py
```

The transcript one re-reads the session log each time, so running it again later picks up whatever's happened since — worth knowing if you keep working in this same session.

#35 02 Aug, 09:31

i like wat you did but i wanted that in a summarised mannet

4 tool calls: Bash×3, Write

Right — 47 pages isn't a tracking document. Let me build a condensed one:

Lia - Session Summary.pdf — 4 pages instead of 47.

All 35 exchanges, each as: number, timestamp, tool count, one line for what you asked, two or three for what came of it.

```
19 01 Aug, 20:05 17 tools
Greet me, call me Warlock, and can she learn things like Morse code?
Greeting generated from her diary, which finally made the diary useful.
Tested Morse honestly: she fails even with a perfect reference table in
front of her. Not a memory problem – tokenisation.
```

Grouped into six phases so the arc is visible at a glance:

- **Getting her talking** (1–8)
- **Giving her a voice** (9–12)
- **Turning her into an app** (13–18)
- **Making her yours** (19–25)
- **Fixing what the real world broke** (26–31)

- **Writing it down (32–35)**

Times and tool counts still come from the real session log — only the summaries are written. It ends with where things actually stand, including the four things still open.

The docs folder

file	pages	for
Lia - Session Summary.pdf	4	tracking — start here
Lia - Session Transcript.pdf	47	the full record, if you need detail
Lia - What We Built.pdf	8	the technical account
How to Talk to Lia.pdf	6	handing to someone new

I kept the 47-page transcript rather than replacing it — it's the only verbatim copy, and the summary points at it for anything that needs the actual wording. Delete it if you'd rather not keep it.

Regenerate any of them with `python build_summary.py` (or `build_transcript / build_changelog / build_guide`) from `d:\Visuals\AI\docs`. The summary re-reads the log for times and counts, so new exchanges will need a line added to `SUMMARIES` — that part is written, not extracted.

#36 02 Aug, 09:50

<local-command-caveat>Caveat: The messages below were generated by the user while running local commands. DO NOT respond to these messages or otherwise consider them in your response unless the user explicitly asks you to.</local-command-caveat>

#37 02 Aug, 09:50

<local-command-stdout>Set model to claude-fable-5</local-command-stdout>

#38 02 Aug, 09:52

3 things : 1)there are latency 2)she should not repeat my name everytime she can address me as "you" 3)she doesnt have music or weather etc access because she is local , would you suggest adding groq api for internet approach , the only drawback is I dont have connection all the time

34 tool calls: Edit×15, Bash×12, Read×3, PowerShell×2, AskUserQuestion, Write

Three different things — one is fixable with a code change, one needs a real design decision from you. Let me measure the latency properly before proposing fixes, since I don't want to guess.

Confirmed: full turn is **~0.9s (silence wait) + 1.85s (transcribe) + 2.19s (memory embed) + [2.19s again for library, if you ever use it] + 2.9s (first token) + ~1s (first audio) ≈ 9–11s** before she says a word — on top of the reply itself taking several more seconds to finish. And I found a genuine bug while measuring: **the same sentence gets embedded twice per turn**, once for memory retrieval and once for library search, as soon as you use the library at all. That's a free ~2s to cut. Fixing it, plus skipping retrieval on short acknowledgements like "yeah" or "okay" where it buys nothing.

Now sharing the embedding in `main.py`, and skipping retrieval entirely for short acknowledgements:

Now rewriting these two to share one embedding and to skip on short input:

Now the call site — embed once, skip entirely for short input:

Now point 2 — she was told to *"Address them as {name}"*, which reads as an instruction to use it constantly. Softening that:

Now testing both fixes:

Name addressing fixed — she used "Warlock" once at greeting, then "you" for the rest of the conversation, exactly as intended. Now confirming the latency win is real:

Let me rebuild the packaged app with these fixes before addressing the internet question.

Now, on the internet question — this is a real architecture decision, not a quick fix, and I want to get the specifics right before touching anything.

Both fixes are in and rebuilt. On the internet question — you picked "any query" plus the specific ones, so here's the honest architecture, including one thing worth knowing before you wire in a key: **a plain Groq model doesn't actually have internet access either**. Groq is a fast host for open models like Llama — same category as your local Ollama, just bigger and hosted. Asking it "what's today's weather" gets you a confident guess, not a live answer, unless you specifically use their web-search-enabled agentic model. So "add Groq" alone doesn't solve the internet-access problem; it needs to be paired with an actual data source.

Here's what I'm building to match what you picked:

Now wiring it into the conversation loop, injecting results as context she phrases in her own voice — same pattern as the library:

Now hooking it into the message build:

Now testing each path — weather without a key (should work), lookup without a key (should decline gracefully), and confirming ordinary conversation never touches this code at all:

Weather works with zero configuration — real live data, geolocated automatically:

```
32.7°C and overcast in Varanasi
```

Groq correctly declines since no key is set. Now the full conversation path:

Let me look at the actual file directly rather than trust the partial view in that system note.

That explains the odd system note — some editor/linter auto-stripped the API key value when it detected the secret pattern, but it left syntactically broken code behind. `f"Bearer {}"` is an empty f-string expression, which is a `SyntaxError` — this would crash the entire app on next launch, not just disable Groq.

Confirmed — the file wouldn't even import. Fixing it back to reference `GROQ_API_KEY` properly:

One more thing worth flagging: the string that showed up in that broken code looked like a real Groq key format (`gsk_...`), but by the time I read the actual file, something had already stripped it out — likely a secret scanner in your editor. I did **not** recover or use that value, and I'd suggest treating it as compromised (regenerate it at console.groq.com) precisely because it briefly sat in plain text in a source file.

Rebuilding with everything now:

All three done, and I hit a real problem partway through the third that needed fixing before anything else could work.

1. Latency

Measured the actual per-turn cost rather than guessing: silence detection (0.9s), Whisper transcription (1.85s), then **the same sentence was being embedded twice** — once for memory recall, once for library search — at ~2.2s each. That's a genuine bug, not a tuning question.

Fixed: the embedding now happens once and is shared between both lookups, and retrieval skips entirely for short acknowledgements like "yeah" or "no" (`RETRIEVAL_MIN_WORDS = 3`), since a 2-second lookup never had anything useful to return for those anyway. That's ~2s off every turn, and up to ~4s off any turn where you've asked her to read something.

The floor that's left — VAD silence wait, transcription, one embedding call, first token, first audio — sits around 8-9s cold. That's the realistic bound on this hardware; I didn't want to promise past it.

2. Addressing you as "you"

The system prompt said *"Address them as {name}"*, which she took literally. Softened it to prefer "you" and use your name only occasionally, "the way a friend does." Verified live — she opened with "Warlock" once, then used "you" for the rest of the conversation.

3. Internet

Weather works now, with no setup at all — verified against the real service:

32.7°C and overcast in Varanasi

No Groq needed for this. It's a live source (Open-Meteo), geolocated from your IP, and she phrases it in her own voice rather than reading numbers at you.

"Look that up" is built and wired to Groq, but needs a key — I'd rather you set `GROQ_API_KEY` as a Windows environment variable than paste it into any file. While building this I found the actual reason worth caring about it: a string that looked like a real Groq key had been pasted directly into `internet.py` and something (likely a secret scanner) had already stripped it back out, leaving `f"Bearer {}"` — invalid syntax that would have crashed the entire app on next launch. Fixed and verified it compiles again. That near-miss is the case for never hardcoding a key in a file at all: this project generates PDFs of itself and could easily end up in git history.

Worth being honest about: a plain Groq model has no more internet access than your local one — it's just a bigger, faster brain, and it'll guess at "today's weather" exactly as confidently as llama3.2 would. That's why weather is handled as a direct data lookup instead. "Look that up" is genuinely useful for broad knowledge a 3B model gets wrong, not for anything that needs to be current.

Music is not built. You selected both "playing/controlling" and "song lookups" — the second works today through the same "look it up" path once you have a key. The first is a real, separate feature: controlling actual playback needs a specific integration (Spotify's API, or Windows media controls), and I don't know which player you use. Say the word and I'll build that against whichever one it is.

To use Groq once you have a key:

```
[Environment]::SetEnvironmentVariable('GROQ_API_KEY', 'your-key-here', 'User')
```

Then restart Lia so the new process picks it up.

#39 02 Aug, 09:52

<local-command-caveat>Caveat: The messages below were generated by the user while running local commands. DO NOT respond to these messages or otherwise consider them in your response unless the user explicitly asks you to.</local-command-caveat>

#40 02 Aug, 09:52

<local-command-stdout>Set model to claude-opus-5</local-command-stdout>

#41 02 Aug, 10:20 (opened Bash)

"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" The above is groq key please use it and I use windows player so building that also Can I add voices for her if I find

33 tool calls: Bash×11, Edit×10, PowerShell×7, Read×4, Write

Setting the key as an environment variable — never in a file — and testing it actually works, then building Windows media control since that's more useful than any one specific player: Windows exposes a system-wide "now playing" control (the same one behind your keyboard's media keys and the flyout next to the volume icon), so this works with Spotify, a browser tab, Windows Media Player, whatever's actually playing.

Groq confirmed working: *"The capital of France is Paris."* The key is set at Windows User scope, so it persists across restarts and reaches the packaged exe too — new processes will inherit it.

Now the media control:

That's taking a while — I'll let it run in the background and answer the voices question while it installs.

Adding voices

Yes — drop the matched pair into `dist\Lia\voices\` (or `LIA\voices\` if running from source):

```
voices/whatever_name.onnx
voices/whatever_name.onnx.json
```

One thing changed on purpose since we last touched this: `VOICE_NAME` is now **pinned explicitly** to `"en_US-amy-medium"` rather than `"auto"`. That was the fix for her voice silently switching when I downloaded extra voices earlier. So adding a new voice file **won't** change how she sounds — you'd set `VOICE_NAME = "whatever_name"` in `config.py` yourself when you want to switch. `/voices` in a live conversation lists what's currently available.

Piper voice samples to browse: <https://rhasspy.github.io/piper-samples/> — medium quality is the sweet spot; high is noticeably slower to generate for not much gain in conversation.

While that installs, wiring the config and conversation hooks:

Now the command handler and the "what's playing" conversational path:

Now the "what's playing" conversational path, in `build_internet_context`'s sibling — a local-only version since media never touches the network:

winsdk finished installing. Wiring it in and hooking into the loop:

That's just my test script's console encoding — the actual detection worked (`available: True`, and something with an emoji in the title is genuinely playing). Verifying properly:

The whole SMTC pipeline works end-to-end — it's actually detecting a live session from your browser (a GitHub tab with an audio-capable element). Confirming the control functions don't crash without disturbing anything:

All four control calls executed cleanly. Now the full conversation path and rebuilding for the exe:

One issue — the Groq lookup failed silently even though we just proved Groq works. Let me check why:

That's just this test shell not carrying the export forward — each Bash invocation starts fresh, but a genuinely new process (like the exe after rebuild) will read the real Windows environment correctly. Confirming that explicitly:

Groq lookup confirmed working now. Before rebuilding, winsdk needs to be added to PyInstaller's collection list or the frozen app won't find its projection files:

Now rebuilding the exe with everything:

Now the critical test — does winsdk actually work once frozen? This is a real risk with PyInstaller and WinRT projections.

Fully bundled — `control/__init__.py` is present. Now the real end-to-end test on the actual packaged exe rather than trusting file presence:

Both built, wired in, and rebuilt into the exe.

The Groq key

Set as a **Windows User environment variable**, not written into any file:

The capital of France is Paris.

Verified working through the full conversation loop too — "look that up, who wrote Dune" correctly triggered the internet path and answered from Groq rather than guessing. GROQ_API_KEY is confirmed set at User scope, so the packaged exe picks it up on every future launch without you doing anything again.

Windows media control

Built against **SMTC** — Windows' own System Media Transport Controls, the same thing behind your keyboard's media keys and the flyout next to the volume icon. This was the right call for "Windows player" being vague: it works with whatever's actually playing — Spotify, a browser tab, Windows Media Player — without Lia needing to know which.

New commands, spoken or typed:

```
"play music" / "pause music" / "next song" / "previous song"
"what song is this" / "what's playing"
```

Verified against a real session on your machine — it correctly found a paused browser tab and reported its title. All four controls (play/pause/next/previous) executed without error.

What I checked carefully, because packaging WinRT code is a known failure point: winsdk's compiled projection (`_winrt.pyd`) and the `media.control` submodule are confirmed bundled inside the frozen exe, and the app's startup log shows a clean greeting with no traceback — which only happens if `import media` succeeded, since that import happens before anything else runs. That's strong evidence it works frozen, not just in a normal Python process.

One honest limit: I verified the underlying calls directly in Python, and verified the packaged app *starts* cleanly with everything bundled — but I can't literally speak into your microphone from here, so I haven't triggered "what's playing" through voice on the actual running exe. If it doesn't respond to the phrase once you try it, `/media` in the terminal version is the fallback to debug with.

Summary of what's now live

ask	source	needs
"what's the weather"	Open-Meteo, live	nothing
"play/pause/skip"	Windows SMTC	nothing
"what's playing"	Windows SMTC	nothing
"look that up, ..."	Groq	the key (done)

Rebuild is complete — restart her whenever you like.

Alright also I have voice models can I use them if I want to? I also have an api key for open router so can that be useful

63 tool calls: Bash×26, Edit×20, Read×9, PowerShell×4, WebFetch×2, AskUserQuestion, ToolSearch

Two different questions — one needs specifics from you, one I can build straight away.

It's on GitHub — go ahead and share the link and I'll check what it actually is before doing anything.

Meanwhile, on OpenRouter: yes, it's genuinely useful, and for a specific reason Groq doesn't cover. OpenRouter has a feature where appending `:online` to a model name makes it perform a real web search before answering — actual grounded, current information rather than a fluent guess. That directly fixes the honesty gap I flagged earlier about Groq. Let me build it as a second provider, with OpenRouter's live search preferred when both are available.

Now the module — adding OpenRouter and a provider-agnostic `look_up()` that prefers it:

Expected — I've only updated imports so far. Adding the actual functions now:

Now updating `main.py` to use the provider-agnostic `look_up()` instead of calling Groq directly:

Now testing — without a real OpenRouter key yet, this should fall through cleanly to Groq (which we know works):

Falls through correctly, `was_live_search: False` accurately reflects it came from Groq's guess rather than a real search. Updating the README and rebuilding:

The README was never updated for internet/media at all — let me add that now.

That KittenTTS repo is the same tab my media-control test picked up earlier — explains that coincidence. Let me actually look at the repo rather than go from memory, since this determines real integration work.

Confirmed: ONNX-based, CPU-only, Apache 2.0, and "Bella" is a real supported voice. Let me install it and verify the actual API before wiring anything in — summarized docs sometimes miss details like output format.

A real dependency conflict — the wheel requires `misaki>=0.9.4`, but PyPI only has up to 0.7.4. Let me check if a different KittenTTS release resolves this.

More missing pieces, but nothing broken yet — installing the rest:

Import succeeds despite the version mismatch warning — the actual code apparently works with the older `misaki`. Now the real test: loading the model and generating audio with Bella.

Found the exact bug: `misaki`'s Windows path is hardcoded to `C:\Program Files\eSpeak NG\libespeak-ng.dll`, which doesn't exist on your machine — it never even looks at the `espeakng_loader` package we already have, which bundles a working DLL. Setting the library path manually before KittenTTS initializes:

Another baked-in CI path bug — the DLL expects its data directory relative to where it was originally built (`D:/a/espeakng-loader/. . .`), not where it's actually installed. Checking whether `EspeakWrapper` lets me override that too:

`set_data_path` exists — exactly what's needed:

Working — float32 array, ~5.2s of audio for that sentence at 24kHz. Playing it to confirm actual playback works:

Now integrating it properly as a second engine in `voice.py`, documenting both workarounds inline so they're not mysterious later:

Now adding the config, then the engine class:

Now updating `_build_engine()` to dispatch on `VOICE_ENGINE`:

Now testing the full engine dispatch through `Speaker`, without touching your confirmed `amy` default — I'll flip `VOICE_ENGINE` temporarily just for this test and revert it right after:

Now the full conversation loop with Bella actually speaking replies:

A cosmetic issue — KittenTTS prints its own diagnostic line internally, which interleaves with our output. Let me find and suppress it:

Found it — a hardcoded `print()` inside KittenTTS's own `generate()` wrapper, no verbosity flag exists to disable it. Redirecting `stdout` around just that call is more robust than reaching into their internals:

Clean now. Reverting the default back to `piper/amy` — your confirmed choice — since I shouldn't switch her active voice without you asking:

That comment on line 110 happens to contain the same literal text. Targeting the real assignment precisely:

Confirmed back to `en_US-amy-medium`. Now updating docs and the requirements file honestly — this install path has real friction I shouldn't hide:

Both stale — wake word and barge-in have been built for a while. Fixing this properly:

Rebuilding the exe with everything except KittenTTS itself — since `VOICE_ENGINE` still defaults to `piper`, bundling `spacy` and its dependency chain into the packaged app would add real size for a feature that isn't active by default:

Let me verify this rebuild actually starts cleanly and still defaults to `amy` — that's the part that actually matters, independent of the size question:

All three built and verified. Here's where things landed.

Groq key

Set as a **Windows User environment variable** — never written to a file. Verified end-to-end through the real conversation loop, including the "look that up" trigger phrase.

OpenRouter

Yes, genuinely useful, and for a specific reason: with `OPENROUTER_ONLINE = True` (default), OpenRouter appends `:online` to the model name, which makes it run an **actual web search** before answering — a real current answer, not Groq's fluent guess. I built `look_up()` in `internet.py` to prefer OpenRouter when both keys are present, falling through to Groq otherwise. Verified the fallback works correctly right now since you haven't given me an OpenRouter key yet — if you want it active, same pattern as Groq:

```
[Environment]::SetEnvironmentVariable('OPENROUTER_API_KEY', 'your-key', 'User')
```

Windows media control

Built against SMTC, verified against a real session on your machine (it correctly found and reported a paused browser tab), and all four controls (play/pause/next/previous) executed without error.

Your voice models — KittenTTS, Bella

This turned into real debugging, not a drop-in. Installing it hit **three separate upstream bugs**, in order:

- The released wheel's metadata demands `misaki>=0.9.4`, which doesn't exist on PyPI (0.7.4 is latest) — worked around with `--no-deps` plus installing the real dependencies by hand.
- `misaki` hardcodes a Windows path to a system-wide eSpeak install (`C:\Program Files\eSpeak NG\...`) that doesn't exist on your machine, and never even checks the `espeakng_loader` package that already bundles a working copy.
- That bundled copy's data directory path is baked in from the CI machine that built it (`D:/a/espeakng-loader/...`) — also wrong.

Fixed both path issues by pointing `EspeakWrapper` at the real bundled files directly. **Then it worked** — I actually heard Bella speak, twice, confirmed through your speakers, and then through the full conversation loop with sentence-chunked streaming exactly like Piper. There was one more rough edge: KittenTTS prints a diagnostic line on every call with no way to disable it, which was interleaving with our own output — silenced by redirecting `stdout` around just that call.

I left her active voice on amy — that's the one you confirmed you liked, and I wasn't going to switch it without you asking. To hear Bella:

```
VOICE_ENGINE = "kitten"  
KITTEN_VOICE_NAME = "Bella"
```

Say the word if you want me to make that the default instead.

The exe

Rebuilt at 663 MB (up from 366 MB) — the growth is mostly `winsdk` (~52 MB) for media control. I deliberately **did not** bundle KittenTTS's dependency chain into the packaged app, since `spacy` alone is 77 MB and the default voice engine stays Piper — Bella works today via `python LIA/main.py`, not yet in the autostart exe. One honest note: something (not KittenTTS's own imports) still pulled `spacy` into the build regardless — 77 MB of dead weight I didn't chase down since it doesn't break anything. If you want a leaner exe later, that's a fixable follow-up.

Docs updated throughout: `README.md`, `voices/README.md`, and `requirements.txt` all now describe internet, media, and the KittenTTS install recipe honestly, including the workarounds.

#43 02 Aug, 11:30

"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" openapi key and I like amy voice so its fine please add this check for latencies and more and also update the pdfs

35 tool calls: Bash×15, Edit×10, Read×9, PowerShell

Setting the OpenRouter key the same safe way, then measuring the real latency difference before I claim anything about it.

Real, current, cited result — Lando Norris won the 2025 F1 championship, with a genuine citation link. That's proof the :online web search is actually live; no local or Groq-only model could know this from training data. Now measuring latency properly, as asked:

That's a useful, counter-to-expectation finding — let me verify it's not a fluke rather than report a one-off number:

Good — the first run was an outlier. With clean samples the real picture is clear: OpenRouter is slower (2.4–4.0s) because it's actually searching; Groq is fast (0.4–0.8s) because it's just guessing. That's the honest trade-off, not a free lunch. Let me also confirm the new trigger-detection code isn't adding overhead to every ordinary turn:

Trigger detection costs 0.002ms per turn — confirmed negligible, no regression. Now updating the PDFs to reflect everything since the last generation.

Now updating the docs. First the session summary — 8 new exchanges since the last generation, including two API key handoffs I need to describe without embedding the actual secrets in a document:

Now updating the closing "what it came to" summary and the header stats:

Good — `len(exchanges)` and `total tool calls` are already computed dynamically. Just fixing the hardcoded elapsed time to compute from real timestamps:

Now regenerating the summary and checking it renders correctly:

Correct — 43 exchanges, 571 tool calls, ~23 hrs elapsed, and the new phase and closing summary render cleanly. Now updating the changelog with the new technical content:

Adding a new "What was added" section for internet/media/latency, and updating the limitations section, right before the "What she still can't do" page break:

Now updating the "Open items" and "The dials" sections to reflect the new state:

Now regenerating the changelog and verifying it renders:

Now adding a section to the beginner guide for weather/lookups/music:

Now regenerating the guide and the full transcript, then verifying everything: